

Accessibility of the unstructured α -tubulin C-terminal tail is controlled by microtubule lattice conformation

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This **fundamental** work reveals that the accessibility of the unstructured C-terminal tails of α - and β -tubulins differs with the state of the microtubule lattice. Their accessibility increases with the expansion of the lattice induced by GTP and certain MAPs, which can then dictate the subsequent interactions between MAPs and microtubules, and post-translational modifications of tubulin tails. The evidence supporting the conclusion is **compelling**, although the characterisation of the probes does not answer whether they directly affect the lattice or expose the C-terminal tails of tubulin. This work will be of great interest to the cytoskeleton field.

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Abstract

Microtubules are cytoskeletal filaments that self-assemble from the protein tubulin, a heterodimer of α -tubulin and β -tubulin, and are important for cell mechanics, migration, and division. Much work has focused on how the nucleotide state of β -tubulin regulates the structure and dynamics of microtubules. In contrast, less is known about the structure and function of the C-terminal tails (CTTs) of α - and β -tubulin which are thought to freely protrude from the surface of the microtubule. To study the CTTs, we developed three different biosensors that bind the tyrosinated α -tubulin CTT (Y- α CTT) on the microtubule lattice. Surprisingly, live imaging of the probes indicates that the Y- α CTT is not accessible under normal cellular conditions. Lattice binding of the Y- α CTT probes can be increased by three different ways of changing the tubulin conformational state: the drug Taxol, expression of microtubule-associated proteins (MAPs) that recognize or promote an expanded tubulin

conformation, or expression of tubulin that cannot hydrolyze GTP. Molecular dynamics simulations indicate that the Y- α CTT undergoes numerous transient interactions with the bodies of α -tubulin and β -tubulin in the lattice, and that the frequency of these interactions is regulated by the tubulin nucleotide state. These findings suggest that accessibility of the Y- α CTT is governed by local nucleotide- and MAP-dependent conformational changes to tubulin subunits within the microtubule lattice.

Impact statement

Using three different biosensors, the authors determine that the C-terminal tail of α -tubulin does not freely extend from the microtubule surface as widely thought, but rather is inaccessible along most microtubules in cells.

Introduction

Microtubules are dynamic polymers that provide mechanical support to cells, serve as tracks for intracellular trafficking, and form the mitotic spindle that separates the replicated genome during cell division (Forth and Kapoor 2017 [↗](#), Prosser and Pelletier 2017 [↗](#), Logan and Menko 2019 [↗](#), Fourriere et al. 2020 [↗](#), Risteski et al. 2021 [↗](#)). Microtubules are formed by self-assembly of the protein tubulin, a heterodimer of α -tubulin and β -tubulin. Although both α - and β -tubulin bind GTP and thus belong to the superfamily of G proteins (Hughes 1983 [↗](#), Nogales et al. 1998 [↗](#)), only β -tubulin hydrolyzes GTP and exchanges GDP for GTP.

The ability of β -tubulin to hydrolyze GTP is intimately linked to microtubule assembly and disassembly (Cleary and Hancock 2021 [↗](#), Gudimchuk and McIntosh 2021 [↗](#), Chew and Cross 2025 [↗](#)). During polymerization, tubulin subunits with GTP bound to β -tubulin (GTP-tubulin) associate in a head-to-tail fashion with the plus (growing) end of the microtubule. After incorporation into the microtubule lattice, β -tubulin hydrolyzes its GTP to GDP, triggering compaction and twist of its associated α -tubulin subunit and a decrease in the longitudinal dimer-to-dimer distance *i.e.* compaction of the lattice (Alushin et al. 2014 [↗](#), Manka and Moores 2018 [↗](#), Zhang et al. 2018 [↗](#), LaFrance et al. 2022 [↗](#)). When GDP-tubulin subunits are exposed at the plus end, microtubules are unstable and depolymerization ensues. Thus, a cap of GTP-tubulin stabilizes the microtubule lattice (**Figure 1A** [↗](#)) (Mitchison and Kirschner 1984 [↗](#)).

Recent work has shown that within the microtubule lattice, GDP-tubulin subunits can be switched from their “compacted” state to a GTP-like “expanded” state by tubulin-binding drugs and microtubule associated proteins (MAPs). Taxol, a natural product that stabilizes microtubules, expands the microtubule lattice *in vitro* and in cells (Alushin et al. 2014 [↗](#), Kellogg et al. 2017 [↗](#), Prota et al. 2023 [↗](#), de Jager et al. 2025 [↗](#)). Similarly, the MAPs kinesin-1, CAMSAP3 and MAP7 can recognize and/or induce an expanded MT lattice (Peet et al. 2018 [↗](#), Shima et al. 2018 [↗](#), Liu and Shima 2023 [↗](#), Shen and Ori-McKenney 2024 [↗](#)). In contrast, the MAPs Tau and MAP2 recognize and/or induce a compacted microtubule lattice (Castle et al. 2020 [↗](#), Siahaan et al.

2022), suggesting that microtubule lattice conformation can be remodeled through MAP binding. Lattice conformation also appears to be altered upon changes in the osmotic environment (Shen and Ori-McKenney 2024 [↗](#)). At least two outcomes have been linked to conformational plasticity of tubulin subunits within the lattice (Verhey and Ohi 2023 [↗](#), Chew and Cross 2025 [↗](#)).

First, molecular motor proteins that step along the microtubule surface can communicate with each other through the lattice (Wijeratne et al. 2022 [↗](#)). Second, expansion of the microtubule lattice licenses tubulin subunits for detyrosination, a tubulin post-translational modification (PTM)

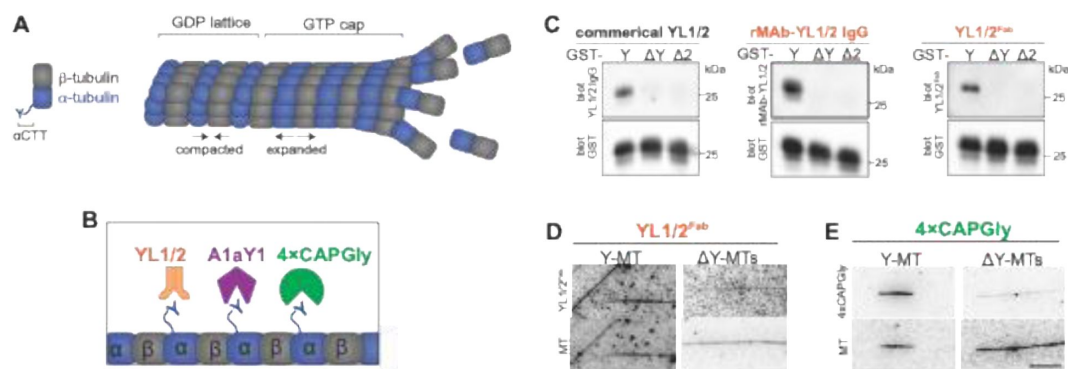


Figure 1.

Three sensors that recognize the Y- α CTT.

(A) Schematic of tubulin states within the microtubule lattice. Tubulin dimers add to the end of a microtubule in a GTP-bound and expanded state, resulting in a stabilizing GTP cap. In the microtubule lattice, β -tubulin undergoes GTP hydrolysis, resulting in a GDP lattice and compaction of α -tubulin. **(B)** Schematic of three sensors (YL1/2, A1aY1, and 4xCAP-Gly) generated to detect the accessibility of the Y- α CTT along the microtubule lattice. **(C)** Validation of the specificity of the YL1/2 antibody-based probes for the Y- α CTT. GST-tagged α CTT sequences were probed by western blotting with the commercial YL1/2 monoclonal antibody, recombinant YL1/2 antibody fused to GFP (rMab-YL1/2-GFP), or a Fab fragment of rMab-YL1/2-GFP (YL1/2^{Fab}-GFP). Each blot was also probed for GST protein as a loading control. Y: full-length and tyrosinated α CTT sequence; Δ Y: detyrosinated α CTT sequence (lacking the C-terminal tyrosine); Δ 2: α CTT sequence lacking the C-terminal two amino acids. **(D,E)** Representative images of (D) YL1/2^{Fab}-GFP or (E) 4xCAPGly-GFP proteins binding to tyrosinated (Y-MT) or detyrosinated (Δ Y-MT) microtubules. HeLa tubulin was polymerized into microtubules and Taxol-stabilized. The microtubules were used directly (Y-MTs) or detyrosinated by VASH/SVBP-containing lysate. Scale bar: 5 μ m.

(Yue et al. 2023 [↗](#)).

A critical element of tubulins is their C-terminal tails (CTTs) (**Figure 1A** [↗](#)), disordered and negatively-charged segments that extend from the body of the tubulin proteins. The CTTs alter microtubule dynamics by impeding microtubule polymerization and increasing the growth-to-shortening transition (Serrano et al. 1984 [↗](#)) and are thought to be a major control hub of the microtubule lattice. The CTTs contain the majority of sequence differences between tubulin isoforms and over 40% of the overall charge of tubulin. They are also hotspots for post-translational modifications (PTMs) that are broadly important for cellular processes including intracellular trafficking, cell division, ciliary beating, neuronal pathfinding, and cardiomyocyte function (Janke and Magiera 2020 [↗](#), Roll-Mecak 2020 [↗](#)). In addition, the CTTs regulate binding and activity of MAPs and molecular motor proteins (Serrano et al. 1984 [↗](#), Serrano et al. 1985 [↗](#), Wang and Sheetz 2000 [↗](#), Niederstrasser et al. 2002 [↗](#), McKenney et al. 2016 [↗](#), Fan and McKenney 2023 [↗](#)). Despite their importance for microtubule biochemistry and function, little is known about the structure or positioning of the CTTs of α - and β -tubulin. Indeed, the CTTs are assumed to freely extend from the microtubule surface.

Interestingly, molecular modeling approaches suggest that the CTTs undergo interactions with the body of tubulin within the lattice (Luchko et al. 2008 [↗](#), Freedman et al. 2011 [↗](#), Laurin et al. 2017 [↗](#), Sahoo and Hanson 2025 [↗](#)) and thus may not always be exposed. We sought to test this idea experimentally and to this end, we developed a set of three structurally-unrelated biosensors that recognize the tyrosinated (native or unmodified) CTT of α -tubulin (Y- α CTT). Strikingly, we find that the Y- α CTT probes show limited binding to microtubules when imaged in live cells, suggesting that the α -tubulin CTT is not accessible along most microtubules in cells. We find that binding of the Y- α CTT probes to microtubules is rapidly increased upon fixation, Taxol treatment, or binding of MAPs that induce an expanded lattice state. We also find that microtubule binding of the Y- α CTT probes is enhanced by GTP-locked tubulins in cells. These findings suggest that accessibility of the α CTT can be locally regulated by changes to the tubulin conformational state.

Results

Biosensors of α CTT accessibility in the microtubule lattice

To examine the accessibility of the α CTT along the microtubule lattice, we developed three probes that recognize the native α CTT sequence of α -tubulin isoforms that contain a C-terminal tyrosine residue (Y- α CTT) (**Figure 1B** [↗](#)). Notably, the three probes are derived from different protein sources, lending rigor and robustness to our results.

We first tested whether the rat monoclonal antibody YL1/2 (Kilmartin et al. 1982 [↗](#), Wehland et al. 1983 [↗](#)) could serve as a sensor of Y- α CTT availability. We determined the sequences of the immunoglobulin G (IgG) heavy and light chains of YL1/2 (**Supp Figure 1A,B** [↗](#)) and then purified an EGFP-tagged recombinant monoclonal antibody (rMAb) version of YL1/2 (rMAb-YL1/2-GFP, **Supp Figure 1C** [↗](#)). To confirm that the experimentally-determined protein sequences generate recombinant protein with efficacy and specificity identical to the original antibody, we tested the ability of rMAb-YL1/2-GFP to recognize the proper antigen using western blotting. We purified glutathione-S-transferase (GST) proteins fused with different human TubA1A α CTT sequences, either full-length [tyrosinated (Y): SVEGEGEEEGEEY], lacking the C-terminal tyrosine [detyrosinated (Δ Y): SVEGEGEEEGEE] or lacking the terminal two amino acids (Δ 2: SVEGEGEEEGE). The rMAb-YL1/2-GFP protein recognized the GST-Y sequence by western blot, but not the GST- Δ Y or GST- Δ C2 sequences, identical to the commercial YL1/2 antibody (**Figure 1C** [↗](#)). To avoid potential issues with bivalency, we then generated a monovalent EGFP-tagged Fragment antibody

binding (Fab) version (**Supp Figure 1A,C**) by papain digestion and confirmed that the YL1/2^{Fab}-GFP protein retains the ability to specifically recognize the GST-Y sequence by western blotting (**Figure 1C**).

We next tested A1aY1, a synthetic protein isolated when screening a yeast display library for proteins that recognize the TubA1A CTT (Kesarwani et al. 2020). We used an A1aY1 probe tagged with superTagRFP [sTagRFP, (Mo et al. 2020)] because a similar construct with TagRFP-T was previously shown to provide strong recognition of the Y- α CTT (Kesarwani et al. 2020). We tested A1aY1 by transient expression in COS-7 cells whose flat morphology enhances visualization of microtubules. We found that sTagRFP-A1aY1 shows little to no microtubule binding in cells that are imaged live (**Supp Figure 2A**) but localizes along microtubules in fixed cells (**Supp Figure 2B,C**). We also found that probe localization was sensitive to expression levels as the sTagRFP-A1aY1 probe was diffusely localized at low levels of expression but could decorate the microtubule lattice at higher levels of expression (not shown). To minimize variability in expression levels, we generated a HeLa-Kyoto stable cell line that expresses sTagRFP-A1aY1 in a doxycycline-inducible manner.

We then tested whether a CAP-Gly domain could act as a sensor of Y- α CTT accessibility based on the known ability of CAP-Gly domains to bind to the Y- α CTT (Honnappa et al. 2006, Peris et al. 2006, Mishima et al. 2007, Weisbrich et al. 2007, Bieling et al. 2008). We generated a construct containing the two tandem CAP-Gly domains and part of the first coiled-coil segment of rat CLIP170 (amino acids 3-484) tagged with mScarlet3 [mSc3, (Gadella et al. 2023)]. The presence of the coiled-coil segment results in a dimeric protein that contains four CAP-Gly domains (hereafter referred to as 4xCAPGly). When transiently expressed in COS-7 cells and imaged live, 4xCAPGly-mSc3 localized to microtubule plus ends (**Supp Figure 3A**) due to the presence of a SxIP motif which enables EB-dependent tip tracking (Bieling et al. 2008). Like A1aY1, fixation caused 4xCAPGly-mSc3 to localize along microtubules (**Supp Figure 3B,C**). We found that 4xCAPGly-mSc3 localization was also sensitive to expression levels as the probe was diffusely localized at low levels of expression but could decorate the microtubule lattice at higher levels of expression (not shown). To minimize variability in expression levels, we generated HeLa-Kyoto stable cell lines that express 4xCAPGly tagged with either mScarlet3 or mEGFP in a doxycycline-inducible manner.

Finally, we tested the specificity of the Y- α CTT probes in reconstitution assays. We purified recombinant versions of the A1aY1 and 4xCAPGly probes from *E. coli*. The recombinant 4xCAPGly-mEGFP protein was soluble (**Supp Figure 1C**), however, the recombinant sTagRFP-A1aY1 probe was prone to aggregation and thus not suitable for reconstitution assays, perhaps due to the oligomeric nature of the sTagRFP moiety (Mo et al. 2020). We also purified tubulin from HeLa S3 cells which are largely devoid of tubulin PTMs including dephosphorylation of the α CTT (Souphron et al. 2019, Thomas et al. 2025). We assembled microtubules and used Taxol to stabilize them in a flow chamber. The Taxol-MTs were either untreated to maintain the tyrosinated state (Y-MTs) or were treated with VASH1/SVBP to generate dephosphorylated microtubules (Δ Y-MTs) (Thomas et al. 2025). Both the YL1/2^{Fab} and 4xCAPGly probes showed stronger decoration of Y-MTs than Δ Y-MTs (**Figure 1D,E**). Taken together, these results demonstrate that we have identified three different probes that specifically recognize the native and unmodified (tyrosinated) α -tubulin CTT.

The α CTT is not accessible within a GDP microtubule lattice

Our initial finding that localization of the A1aY1 and 4xCAPGly probes shifts from cytosolic (A1aY1) or microtubule plus end (4xCAPGly) to the microtubule lattice upon fixation (**Supp Figures 2** and **3**) suggests that lattice state impacts the accessibility of the Y- α CTT. Previous work showed that localization of A1aY1 shifts from cytosolic or plus ends to the microtubule lattice in cultured cells treated with SiR-Tubulin (Kesarwani et al. 2020), a Taxol derivative that can convert GDP-tubulin into a GTP-like expanded state (Siahaan et al. 2022). We thus hypothesized that lattice expansion drives accessibility of the Y- α CTT.

To test this idea, we used live-cell imaging to examine the localization of the Y- α CTT probes before and after Taxol treatment. Given the requirements of live-cell imaging and low probe expression (**Supp Figures 2** and **3**), we utilized the stable HeLa-Kyoto cell lines to inducibly express low levels of the probes sTagRFP-A1aY1 (A1aY1 hereafter for brevity) or 4xCAPGly-mEGFP (4xCAPGly hereafter for brevity). In untreated cells, the A1aY1 probe localized diffusely throughout the cell with faint localization along the microtubule lattice (**Figure 2A, before**). No change was observed upon addition of DMSO vehicle control (**Figure 2A,B**).

Strikingly, within minutes of Taxol addition, the A1aY1 probe localization to the microtubule lattice was significantly increased (**Figure 2A,C,D**), suggesting that the Y- α CTT became accessible to the probe after Taxol-induced expansion of the microtubule lattice. To verify these results in another cell line, we transiently expressed the A1aY1 probe in HeLa and COS-7 cells and found a similar increase in microtubule lattice decoration after Taxol treatment (**Supp Figure 4**).

We carried out similar experiments to examine the response of the 4xCAPGly probe to Taxol treatment. In untreated cells, the 4xCAPGly probe localized to the growing plus ends of microtubules via its association with endogenous EB proteins (**Figure 2E, before**), consistent with (Bieling et al. 2008). No change was observed upon addition of DMSO vehicle control (**Figure 2E,F**). Within minutes of Taxol treatment, the 4xCAPGly probe relocated from the ends to the lattice of the microtubules (**Figure 2E,G,H**), suggesting that Taxol-mediated expansion of the microtubule lattice renders the Y- α CTT accessible for 4xCAPGly probe binding. Similar results were obtained when the 4xCAPGly probe was transiently expressed in HeLa or COS-7 cells (**Supp Figure 4**).

Taken together, these results suggest that the Y- α CTT is minimally accessible when tubulin in the microtubule lattice is in the GDP and compacted state but is exposed and accessible for probe binding when GDP-tubulin subunits are pharmacologically converted into a GTP-like expanded state.

Lattice expansion by MAPs causes increased accessibility and modification of the α CTT

In cells, GDP-tubulin can be converted into a GTP-like expanded state by binding of specific MAPs to the microtubule lattice. We thus tested whether MAP binding can change the ability of the Y- α CTT probes to recognize the microtubule lattice. We transfected the stable A1aY1 and 4xCAPGly cell lines with four different MAPs known to recognize and/or induce an expanded state: a rigor version of kinesin-1 (KIF5C^{rigor}), CAMSAP2, CAMSAP3, and MAP7 (Peet et al. 2018, Shima et al. 2018, Liu and Shima 2023, Shen and Ori-McKenney 2024). As controls, we expressed the MAPs tau and MAP2 that recognize and/or induce a compacted lattice state (Castle et al. 2020, Siahhaan et al. 2022). Expression of MAPs that expand the microtubule lattice (KIF5C^{rigor}, CAMSAP2, CAMSAP3, or MAP7) resulted in colocalization of the A1aY1 probe with the expanded lattice (**Figure 3B**) whereas expression of MAPs that compact the microtubule lattice (tau or MAP2) did not change the localization of the A1aY1 or 4xCAPGly probes as compared to the control conditions (untransfected or mEGFP-tagged human TubA1A expression) (**Figure 3A,C**). Similar results were found with the 4xCAPGly probe: only expression of expander MAPs resulted in a significant increase in 4xCAPGly probe binding along the microtubule lattice (**Figure 3B,D**). These results indicate that in cells, MAPs that expand the lattice expose the α -tubulin CTT whereas MAPs that compact the lattice do not (**Figure 3E**). We suggest that MAP binding drives a local expansion of the underlying microtubule lattice that results in switching of the α CTT into an accessible state recognized by the Y- α CTT probes.

The ability of MAPs to regulate Y- α CTT accessibility suggests that they could regulate downstream events such as post-translational modification of the α CTT. Consistent with this possibility, the expander MAPs CAMSAP2 and CAMSAP3 enable detyrosination (ΔY) of α -tubulin subunits by the

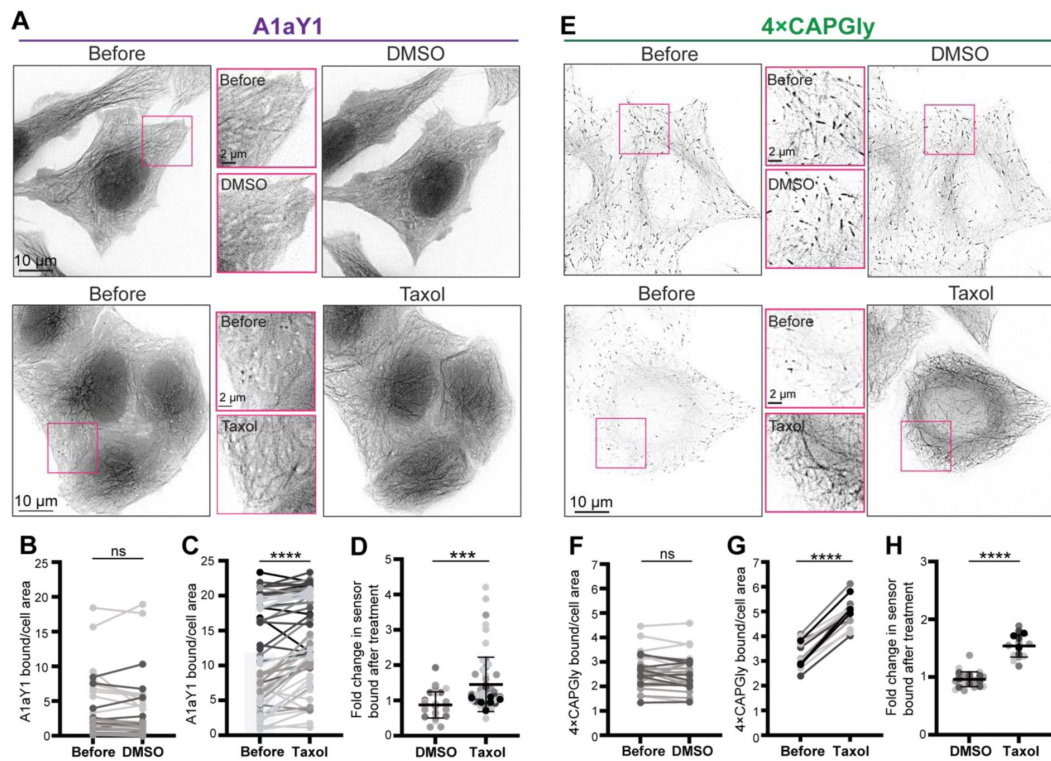


Figure 2.

Y-αCTT probes bind to the microtubule lattice after Taxol treatment.

(A-D) Live-cell imaging of A1aY1 probe. **(A)** Representative images of the stable sTagRFP-A1aY1 HeLa cell line before and 15 minutes after treatment with (top) 0.3% DMSO control or (bottom) 10 μ M Taxol. Insets (boxes) show magnified views of A1aY1 probe. Scale bars, 10 μ m. **(B,C)** Quantification of probe binding to microtubules. Paired data plots display the localization of A1aY1 probe in individual cells before and after treatment with (B) DMSO or (C) Taxol. **(D)** Mean difference plot showing the fold change in A1aY1 probe binding in DMSO-vs Taxol-treated cells. DMSO=29 cells across 3 experiments; Taxol=59 cells across 7 experiments. **(E-H)** Live-cell imaging of 4xCAPGly probe. **(E)** Representative images of the stable 4xCAPGly-mEGFP HeLa cell line before and 15 minutes after treatment with (top) 0.3% DMSO control or (bottom) 10 μ M Taxol. Insets (boxes) show magnified views of CAPGly probe. Scale bars, 10 μ m. **(F,G)** Quantification of probe binding to microtubules. Paired data plots display the localization of 4xCAPGly probe in individual cells before and after treatment with (F) DMSO or (G) Taxol. **(H)** Mean difference plot showing the fold change in 4xCAPGly probe binding in DMSO- or Taxol-treated cells. DMSO=28 cells across 3 experiments; Taxol=17 cells across 5 experiments. Error bars in (D) and (H) indicate SD. ***: $p < 0.0002$; ****: $P < 0.0001$; ns: not significant [Student's t test (B,C,F,G: two-tailed; paired), (D,G: unpaired)].

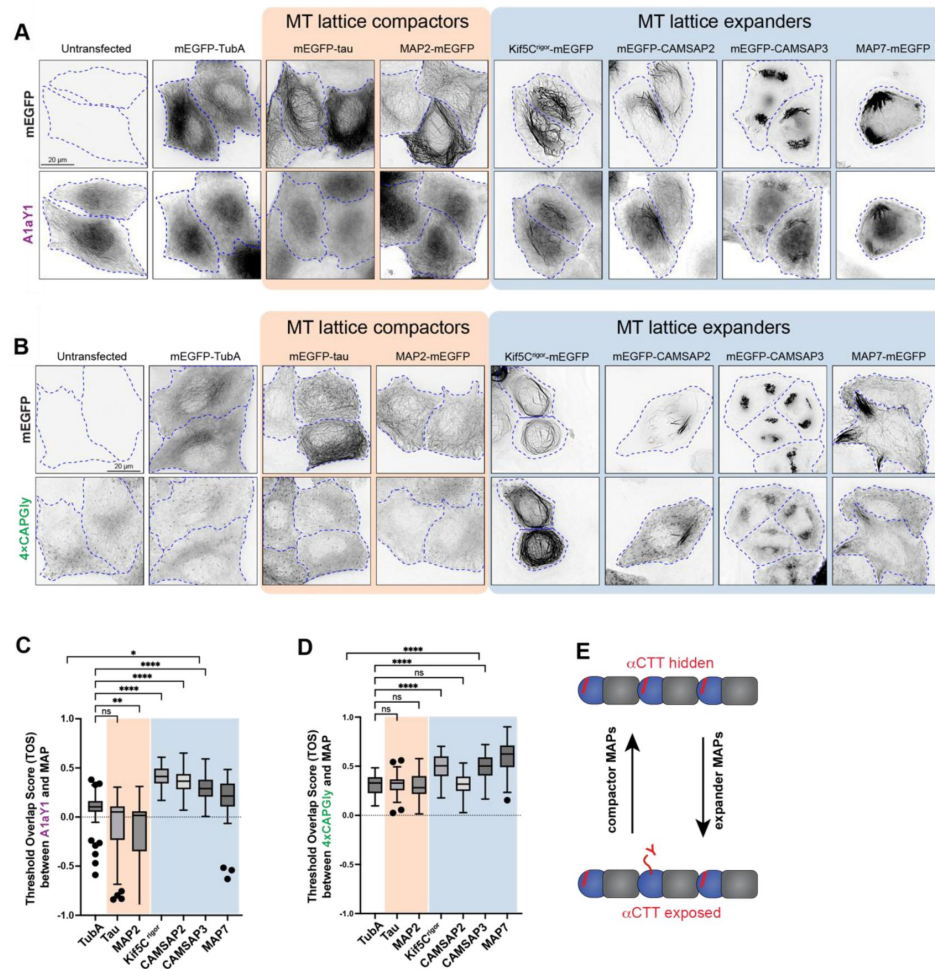


Figure 3.

MAPs that expand the microtubule lattice increase Y- α CTT probe binding.

(A,B) Representative live-cell images of stable (A) A1aY1 or (B) 4xCAPGly HeLa cell lines transiently expressing the indicated GFP-tagged tubulin or MAP constructs. Cell boundaries are indicated by blue dotted lines. Scale bars, 20 μ m. (C,D) Quantification of (C) A1aY1 or (D) CAPGly probe colocalization with GFP-tagged tubulin or MAP constructs. The threshold overlap score (TOS) was measured on a per-cell basis where 1.0 indicates perfect colocalization, -1.0 indicates perfect anti-colocalization, and values near 0 indicate no relationship. Data from three independent experiments are presented as Tukey box plots. The box encompasses the 25th to 75th percentiles, with a line at the median. Whiskers show the last data point within 1.5 times the interquartile range. Outliers are plotted as individual points. *: $p < 0.1$; **: $p < 0.001$; ****: $p < 0.0001$; ns: not significant (Kruskal-Wallis test followed by post-hoc Dunn's multiple pairwise comparisons with TubA1A as the control). Number of cells analyzed (n) in (C): TubA1A = 69, Tau = 57, MAP2 = 66, Kif5C^{rigor} = 70, CAMSAP2 = 50, CAMSAP3 = 56, and MAP7 = 55 and in (D): TubA1A = 62, Tau = 61, MAP2 = 61, Kif5C^{rigor} = 76, CAMSAP2 = 62, CAMSAP3 = 71, and MAP7 = 57. (E) Schematic model depicting how expander and compactor MAPs regulate microtubule lattice conformation, influencing Y- α CTT accessibility.

enzyme VASH1/SVBP (Yue et al. 2023 [\[1\]](#)). To extend these findings, we tested whether the same expander and compactor MAPs that regulate Y- α CTT accessibility in cells (Figure 3 [\[2\]](#)) cause an increase in Δ Y-MTs. Indeed, like CAMSAP2 and CAMSAP3 (Yue et al. 2023 [\[1\]](#)), expression of the expander MAPs KIF5C^{rigor} or MAP7 resulted in significant increase in Δ Y-MTs (Figure 4 [\[3\]](#)). In contrast, expression of the compactor MAPs tau or MAP2 did not cause an increase in Δ Y-MTs (Figure 4 [\[3\]](#)).

Nucleotide state regulates α CTT accessibility and detyrosination

We next tested whether the GTP-vs GDP-state of tubulin can regulate accessibility of the α CTT. To test this in the stable A1aY1 and 4xCAPGly cell lines, we generated a plasmid that expresses PA-tagged human TubA1A with an internal ribosome entry site (IRES) driving the expression of an EGFP reporter protein. We compared WT TubA1A to the mutant E254A which prevents GTP hydrolysis by β -tubulin and therefore locks tubulin in a GTP-bound state (Roostalu et al. 2020 [\[4\]](#), LaFrance et al. 2022 [\[5\]](#), Beckett and Voth 2023 [\[6\]](#)).

We transfected the stable A1aY1 and 4xCAPGly cell lines with plasmids for expressing TubA1A(WT) or TubA1A(E254A) and examined the localization of the Y- α CTT probes by live-cell microscopy. In cells expressing WT TubA1A (visualized by expression of the EGFP reporter), the A1aY1 probe localized in a diffuse manner whereas in cells expressing TubA1A(E254A), the A1aY1 probe localized along the microtubule lattice (Figure 5A [\[7\]](#)). Similar results were found for the 4xCAPGly probe which localized to microtubule plus ends in cells expressing TubA1A(WT) but localized along the lattice in cells expressing TubA1A(E254A) (Figure 5C [\[8\]](#)). Quantification of these results confirmed that expression of GTP-locked E254A α -tubulin resulted in a significant increase in the density of both A1aY1 and 4xCAP-Gly probes along the microtubule lattice (Figure 5B,D [\[9\]](#)). We then examined whether GTP-locked tubulin could also cause an increase in Δ Y-MTs. We found that expression of GTP-locked E254A α -tubulin but not WT α -tubulin resulted in an increase in Δ Y-tubulin levels as assessed by immunostaining (Figures 5E,F [\[10\]](#)) and immunoblotting (Supp Fig 5 [\[11\]](#)). These results suggest that the nucleotide state of tubulin regulates Y- α CTT accessibility for probe binding and PTM enzymes.

MAPs but not nucleotide state regulate α CTT accessibility in vitro

We then used reconstitution experiments to directly probe the link between nucleotide state and accessibility of the α CTT. We polymerized HeLa S3 tubulin in the presence of the GTP analog guanosine-5'-[(α,β -methylene)tri-phosphate (GMPCPP) to generate microtubules in the GTP-like expanded state (GMPCPP-MTs) or in the presence of GTP which is hydrolyzed to generate the GDP compacted state (GDP-MTs). Furthermore, to directly compare Y- α CTT probe binding between GMPCPP-MTs and GDP-MTs in the same flow chamber, we labeled GMPCPP-MTs with fluorescent Alexa Fluor 568-labeled tubulin and GDP-MTs with fluorescent Alexa Fluor 647-labeled tubulin. We mixed the two microtubule populations with purified Y- α CTT probe proteins (Supp Figure 1C [\[12\]](#)) in the same chamber and stabilized the microtubules against depolymerization with 25% glycerol. We found that the purified 4xCAPGly probe bound to both microtubule populations (Figure 6A [\[13\]](#)) with no significant difference in steady-state binding (Figure 6B [\[14\]](#)). Similar results were obtained for the YL1/2^{Fab} which bound to both GMPCPP- and GDP-MTs in the flow chamber (Figure 6C [\[15\]](#)) with no significant difference in steady-state binding (Figure 6D [\[16\]](#)). Thus, although nucleotide state can directly influence tubulin conformational state in cryoEM structures of *in vitro* polymerized microtubules (Alushin et al. 2014 [\[17\]](#), Manka and Moores 2018 [\[18\]](#), Zhang et al. 2018 [\[19\]](#), LaFrance et al. 2022 [\[5\]](#)), it does not appear to directly regulate accessibility of the α -tubulin CTT within the microtubule lattice.

The inability of the *in vitro* polymerized microtubules (Figure 6A-D [\[13\]](#)) to recapitulate the influence of nucleotide state on α CTT accessibility in cells (Figure 5A-D [\[7\]](#)) suggests that the *in vitro* assays are missing key factors that regulate α CTT accessibility. We hypothesized that MAPs

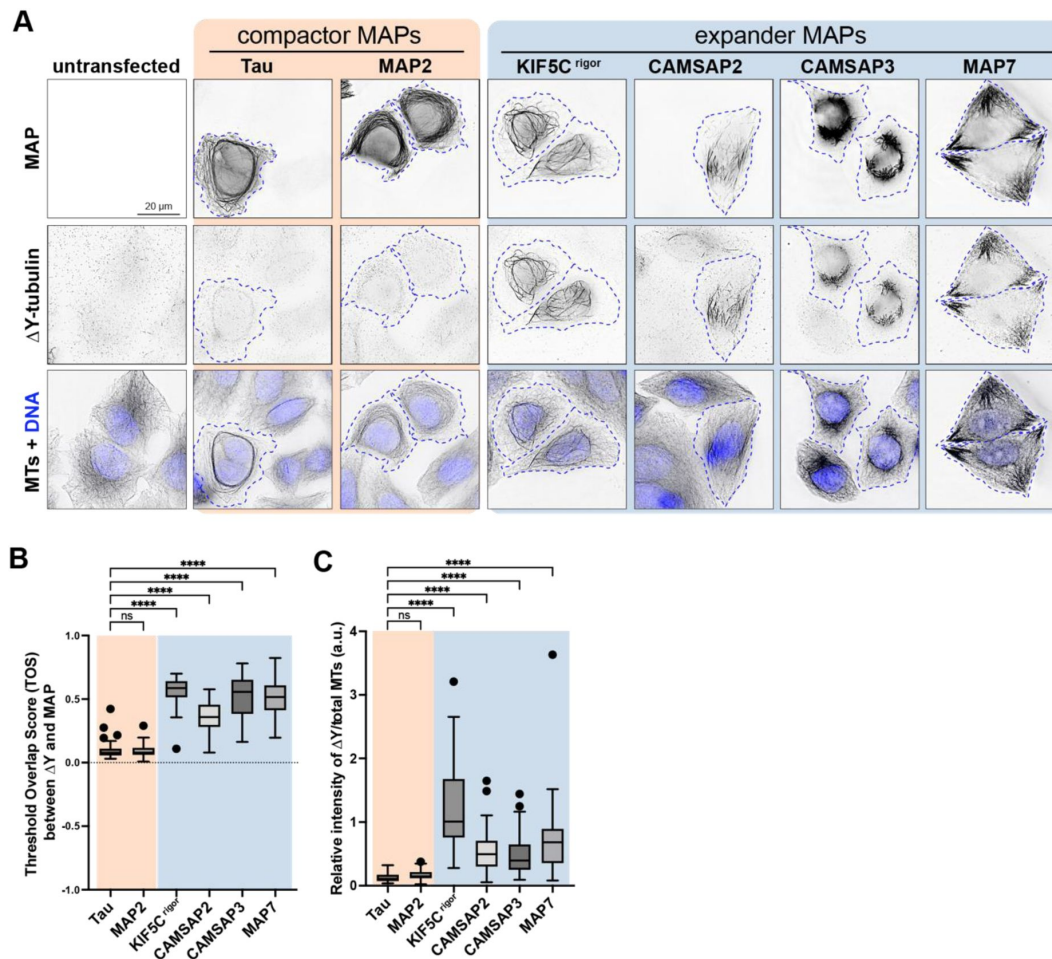


Figure 4.

MAPs that expand the MT lattice increase deetyrosination of the Y- α CTT.

(A) Representative images of HeLa cells transiently expressing the indicated GFP-tagged MAPs and then fixed and stained with antibodies against deetyrosinated microtubules (Δ Y-tubulin) and total microtubules (MTs). Images are shown in inverted grayscale. The nuclei are represented by blue pseudocolor in the bottom panels. Blue dotted lines: boundaries of cells expressing the corresponding MAPs. Scale bar, 20 μ m. **(B)** Quantification of the colocalization of MAPs and deetyrosinated microtubules. The threshold overlap score (TOS) was measured on a per-cell basis. Data from three independent experiments are presented as Tukey box plots. The box encompasses the 25th to 75th percentiles, with a line at the median. Whiskers show the last data point within 1.5 times the interquartile range. Outliers are plotted as individual points. ****: $p < 0.0001$; ns: not significant (Kruskal-Wallis test followed by post-hoc Dunn's multiple pairwise comparisons with tau as the control). Number of cells analyzed (n): Tau = 70, MAP2 = 72, Kif5C^{rigor} = 76, CAMSAP2 = 66, CAMSAP3 = 69, and MAP7 = 67. **(C)** Quantification of the intensity of deetyrosination on MAP-bound microtubules. The fluorescence intensity of deetyrosination was measured on MAP-decorated microtubules and normalized against the total microtubule intensity of MAP-decorated microtubules. Data from three independent experiments are presented as Tukey box plots. ****: $p < 0.0001$; ns: not significant (Kruskal-Wallis test followed by post-hoc Dunn's multiple pairwise comparisons with tau as the control). Number of cells analyzed (n): Tau = 67, MAP2 = 70, Kif5C^{rigor} = 75, CAMSAP2 = 67, CAMSAP3 = 70, and MAP7 = 67.

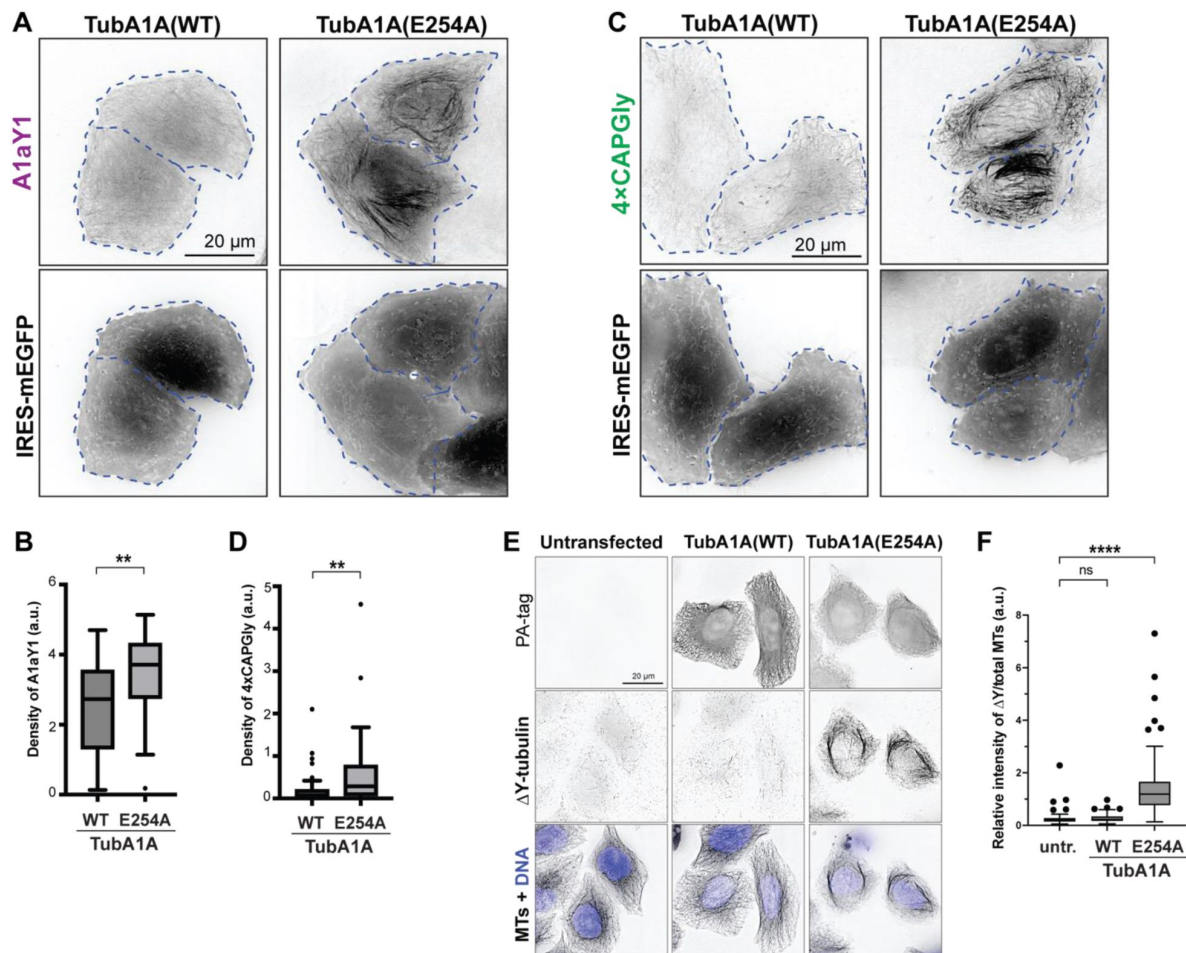


Figure 5.

GTP-like tubulin state increases Y- α CTT accessibility and detyrosination.

(A-D) Live-cell imaging of Y- α CTT probes. **(A,C)** Representative images of stable (A) A1aY1 or (C) 4xCAPGly HeLa cell lines expressing PA-tagged WT or E254A α -tubulin with an IRES-driven GFP protein as a reporter of transfected cells. Cell boundaries are indicated by blue dotted lines. Scale bars, 20 μ m. **(B,D)** Quantification of (B) A1aY1 or (D) 4xCAPGly probe binding to microtubules. The density was measured as the ratio of the skeletonized probe-decorated microtubule length to the total cell area. Data from three independent experiments are presented as Tukey box plots. The box encompasses the 25th to 75th percentiles, with a line at the median. Whiskers show the last data point within 1.5 times the interquartile range. Outliers are plotted as individual points. **: $p < 0.01$ (Mann-Whitney U test). Number of cells analyzed (n) in (B): WT = 33, E254A = 29 and in (D): WT = 42, E254A = 50. **(E,F)** Detyrosinated microtubules. **(E)** Representative images of HeLa cells transiently expressing PA-tagged WT or E254A α -tubulin and then fixed and stained with antibodies against the PA tag, detyrosinated microtubules (Δ Y-tubulin), and total microtubules (MTs). Images are shown in inverted grayscale. The nuclei are represented by blue pseudocolor in the bottom panels. Scale bar, 20 μ m. **(F)** Quantification of the intensity of detyrosination in cells expressing PA-tagged WT or E254A α -tubulin. The fluorescence intensity of detyrosination was measured on a per-cell basis and normalized against the total microtubule intensity. Data from three independent experiments are presented as Tukey box plots. ****: $p < 0.0001$; ns: not significant (Kruskal-Wallis test followed by post-hoc Dunn's multiple pairwise comparisons with the untransfected sample (untr.) as the control). Number of cells analyzed (n): untransfected = 105, WT = 84, E254A = 94.

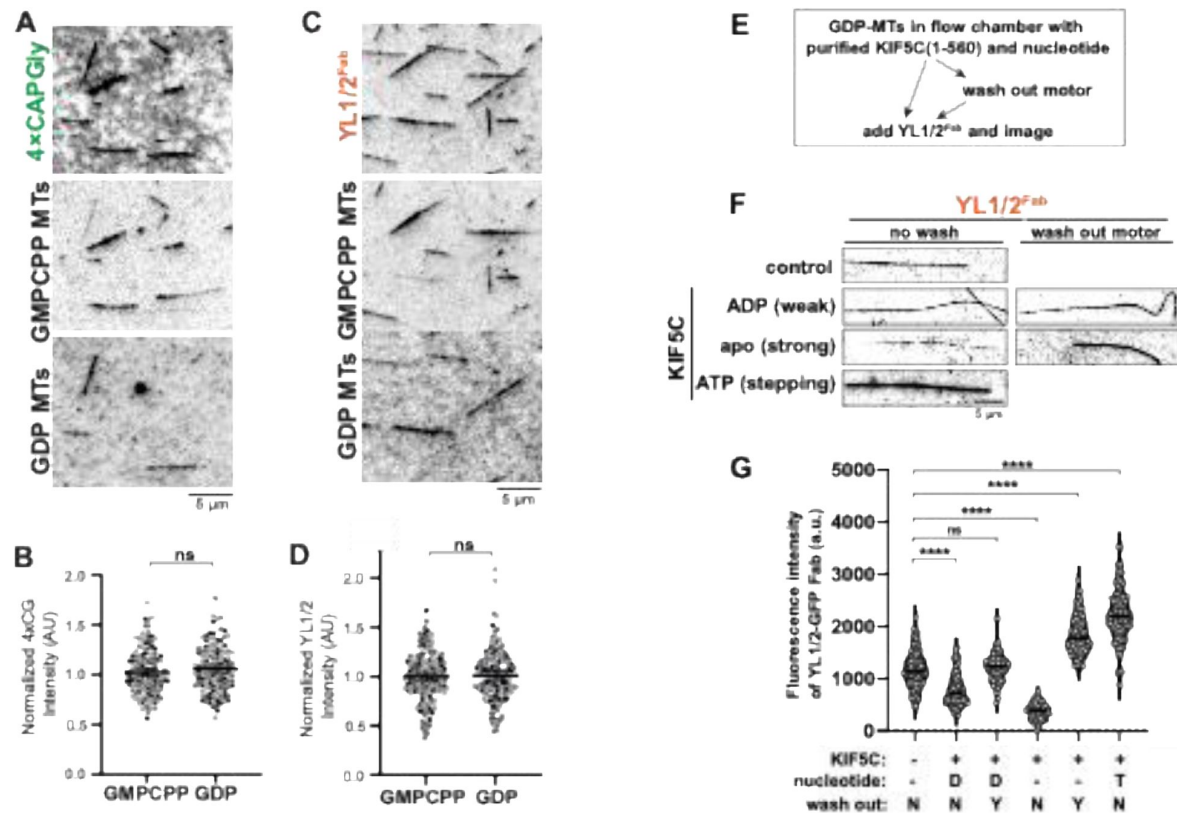


Figure 6.

MAPs but not nucleotide state promote Y-αCTT exposure in vitro.

(A-D) Nucleotide state does not determine probe binding. **(A,C)** Representative images of (A) 4xCAPGly-mEGFP or (C) YL1/2^{Fab}-GFP binding to a mixture containing both AlexaFluor-568 labeled GMPCPP-stabilized microtubules and AlexaFluor-647 labeled GDP microtubules. **(B,D)** Quantification of the fluorescence intensity of (B) 4xCAPGly-mEGFP or (D) YL1/2^{Fab}-GFP probe binding per length of microtubule. Data are presented as scatter plots with data from three independent experiments in different shades of gray. ns: not significant (two-tailed, Student's *t* test). **(E-G)** Stepping KIF5C can increase YL1/2^{Fab} probe binding. **(E)** Flowchart of the in vitro reconstitution assay examining the effect of KIF5C in different nucleotide states on YL1/2^{Fab} binding. **(F)** Representative images of YL1/2^{Fab}-GFP probe binding to GDP-MTs in the absence or presence of KIF5C in different nucleotide states. **(G)** Quantification of the mean fluorescence intensity of YL1/2^{Fab}-GFP probe along GDP-MTs under the conditions shown in (F). Each spot indicates the probe binding on an individual microtubule. Number of microtubules (*n*) = 74-105 from three independent experiments. n.s., not significant, *****p*<0.0001 (two-tailed, *t*-test).

that recognize the nucleotide and/or conformational state of the microtubule lattice are key to regulating α CTT accessibility. To test this hypothesis in the reconstitution assay, we tested whether the kinesin-1 motor KIF5C could regulate Y- α CTT probe binding. For this, we combined GDP-MTs in a flow chamber with purified KIF5C(1-560) protein and the YL1/2^{Fab} (**Figure 6E, left arrow** [↗](#)). The ability of KIF5C(1-560) to interact with microtubules and modulate the lattice state can be regulated by nucleotide. In ADP, KIF5C(1-560) binds weakly to microtubules whereas in the absence of nucleotide (apo state), it is locked in a strong MT-bound state that can induce tubulin expansion. In the presence of ATP, KIF5C(1-560) steps along the lattice and induces transient changes in tubulin conformation ([Peet et al. 2018](#) [↗](#), [Shima et al. 2018](#) [↗](#)).

When KIF5C(1-560) was added in the presence of ADP, the binding of YL1/2^{Fab} to GDP-MTs was reduced as compared to the control condition (no KIF5C) (**Figure 6F,G** [↗](#)). Binding of the YL1/2^{Fab} was even further reduced when KIF5C(1-560) was strongly bound to the GDP-MTs in the apo (no nucleotide) state (**Figure 6F,G** [↗](#)). In contrast, stepping of KIF5C(1-560) in the presence of ATP resulted in a significant increase in YL1/2^{Fab} binding (**Figure 6F,G** [↗](#)).

We considered the possibility that Y- α CTT probe binding may be affected by KIF5C, *e.g.*, through steric hindrance. We thus separated the KIF5C(1-560) microtubule binding from the YL1/2^{Fab} microtubule binding using a washing step (**Figure 6E, right arrows** [↗](#)). After washing out weakly-bound KIF5C(1-560) (ADP condition), YL1/2^{Fab} binding to GDP-MTs was similar to that of the control (no KIF5C) (**Figure 6F,G** [↗](#)). After washing out strongly-bound KIF5C(1-560) (apo state), YL1/2^{Fab} binding to GDP-MTs was significantly increased compared to the control (no KIF5C) or weakly-bound KIF5C(1-560) (ADP state) (**Figure 6F,G** [↗](#)). These results suggest that strong binding of KIF5C(1-560) to the microtubule lattice, either statically in the apo state or transiently in the ATP state, results in increased Y- α CTT probe binding.

Molecular Dynamics simulations show numerous transient interactions of the α CTT tail with the GDP microtubule lattice

To probe the accessibility of the α CTT at an atomistic level, we performed molecular dynamics (MD) simulations of microtubules in the GDP state. The simulations identified four distinct sites where the acidic residues of the α CTT form salt bridges with basic residues in the tubulin body (**Figure 7A** [↗](#), **Supp Figure 6** [↗](#)). Two of these sites are on the body of β -tubulin of the adjacent tubulin along the same protofilament (towards the minus end of the microtubule) [**Figure 7A, site 1 (green) and site 2 (cyan)** [↗](#)] whereas the other two sites are on the body of the α -tubulin containing that CTT [**Figure 7A, site 3 (purple) and site 4 (salmon)** [↗](#)]. Since there are seven glutamates in the α CTT and five basic residues in binding sites 2, 3 and 4, there is a very large multiplicity of interacting conformations between the α CTT and the microtubule (**Supp Movie 1** [↗](#)).

We next performed MD simulations of microtubules in the GTP state and found that the α CTT formed electrostatic interactions with the same four binding sites in the tubulin body (**Figure 7A** [↗](#)). In both GDP and GTP microtubules, interactions between the α CTT and the tubulin body occurred most frequently between the N-terminal part of the α CTT (E441 and E443) and site 1 of the adjacent β -tubulin (**Figure 7B** [↗](#)) whereas fewer interactions were observed between the more C-terminal residues of the α CTT (residues 445-450) and the four binding sites (**Figure 7B** [↗](#)). However, the interactions of the more C-terminal part of the α CTT were more susceptible to the nucleotide state of the lattice. In particular, interactions between residues 445-450 of the α CTT and site 2 of the adjacent β -tubulin were readily observed in the GDP microtubule but nearly abolished in the GTP microtubule (**Figure 7B** [↗](#)). Likewise, cis interactions between residues 449 and 450 of the α CTT with site 4 of the α -tubulin body were observed in the GDP microtubules but not the GTP microtubules (**Figure 7B** [↗](#)).

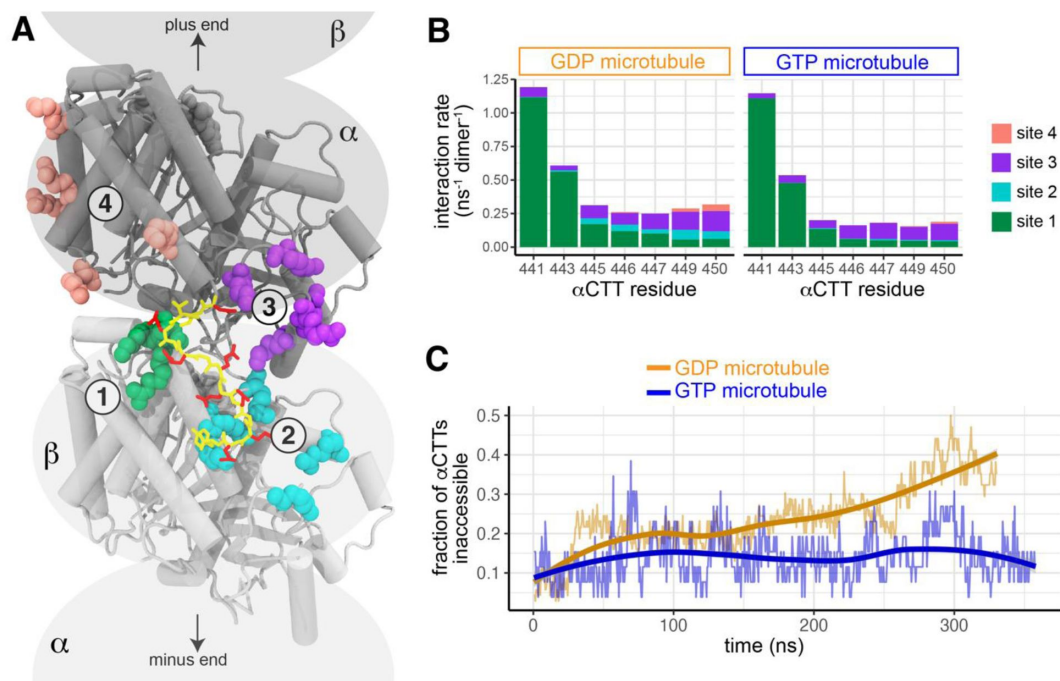


Figure 7

The nucleotide state alters Y-αCTT interactions with the microtubule body.

(A) Representative image from MD simulations identifying four distinct sites where the Y-αCTT interacts with the body of tubulin subunits in the microtubule: sites 1 (green) and 2 (cyan) are on the adjacent β-tubulin along a protofilament (i.e. next tubulin towards the microtubule minus end) whereas sites 3 (purple) and 4 (salmon) are cis-interactions with α-tubulin itself. The tubulin body is shown in cartoon and colored gray. The Y-αCTT is shown in stick and colored yellow with the glutamate sidechains in red. **(B)** Interaction rate of the glutamate residues in the Y-αCTT with the four tubulin body sites for microtubules in the (gold) GDP or (blue) GTP states. **(C)** The fraction of Y-αCTTs that are inaccessible as a function of time for microtubules in the (gold) GDP or (blue) GTP state where inaccessibility is defined as one or more salt bridges formed between glutamates E445-E450 and the interaction sites in the microtubule body.

We hypothesized that the decreased interactions between the α CTT and the tubulin body in the GTP microtubule would render the α CTT more accessible when tubulin is in the GTP and expanded state. To test this, we calculated how many C-terminal residues of the tail (E445 to Y451) were inaccessible (*i.e.* bound to one of the tubulin body sites) throughout the course of the simulations. We found that the α CTT residues in the GDP microtubule showed increasing interactions with the tubulin body as a function of time and reached more than 40% inaccessibility after about 300 ns (**Figure 7C**). In contrast, the α CTTs in the GTP microtubule were only about 10% inaccessible throughout the entire simulation time (**Figure 7C**). Taken together, these results support the hypothesis that the α CTT undergoes frequent interactions with the GDP-containing microtubule lattice that render it inaccessible to the Y- α CTT probes.

Discussion

The α CTT is more accessible along expanded microtubules

In this work, we sought to gain an understanding of the relative accessibility of the α CTT along the microtubule. To this end, we developed three probes to visualize the tyrosinated α CTT: YL1/2^{Fab}, A1aY1, and 4xCAPGly. We note that our probes are not expected to recognize the two α -tubulin isoforms that do not have a genetically encoded C-terminal tyrosine, TubA4A and TubA8, and play roles in brain development and spermatogenesis, respectively (Diggle et al. 2017, Hausrat et al. 2021, Benkirane et al. 2024). The YL1/2^{Fab} and 4xCAPGly proteins can be purified for reconstitution assays whereas the A1aY1 and 4xCAPGly probes can be expressed in cells although their optimal use requires live-cell imaging. It is possible that fixation causes the microtubule lattice to expand, but we cannot exclude alternative explanations, *e.g.*, that fixation disrupts CTT-binding sites on the tubulin body. We also found that high levels of Y- α CTT probe expression can drive their binding to the microtubule lattice, and it is thus critical to limit examination to cells that express low levels of our probes. This is an important caveat to keep in mind in experiments that utilize transient transfection of these Y- α CTT probes.

We found that the A1aY1 and 4xCAPGly probes showed minimal binding to the microtubule lattice in cells under normal culture conditions. This finding suggests that the α CTT is generally not accessible along the microtubule lattice. Faint microtubule localization is more commonly observed with A1aY1 when compared to 4xCAPGly, which may be due to higher affinity interactions of 4xCAPGly with EB1 and other proteins at the microtubule plus end versus the microtubule lattice. The weak lattice binding of the α CTT probes was surprising as the tubulin CTTs are thought to freely protrude from the microtubule lattice.

Expansion of the microtubule lattice via three orthogonal approaches resulted in an increase in Y- α CTT probe binding: 1) treatment of cells with Taxol (**Figure 2**), 2) expression of MAPs that expand tubulin within the microtubule lattice (expander MAPs) (**Figure 3**), and 3) expression of a GTP-locked α -tubulin mutant (E254A) (**Figure 5**). These results suggest that accessibility of the α CTT is regulated by the compacted versus expanded state of tubulins within the microtubule lattice. We note that “expanded” and “compacted” are used as terms relative to one another rather than defined against a standard microtubule structure (LaFrance et al. 2022).

The simplest mechanistic explanation for how tubulin conformational state regulates α CTT accessibility is that the nucleotide state gates α CTT accessibility. However, we find that YL1/2^{Fab} and 4xCAPGly probes bind similarly to GMPCPP-MTs vs GDP-MTs polymerized *in vitro* (**Figure 5**). At first glance, these data conflict with our observation that VASH1/SVBP preferentially detyrosinates microtubules that are in an expanded state (Yue et al. 2023). However, it is possible that an expanded lattice promotes activation of VASH1/SVBP, or, alternatively, that engagement of the α CTT with the catalytic site of VASH1/SVBP is optimal on expanded microtubules. Future work is necessary to address these interesting possibilities.

There are several explanations for why α CTT accessibility in reconstitution assays (**Figure 6A-D**) does not correlate with the structural state of the microtubule lattice as observed in cells (**Figure 5**). First, microtubule assembly *in vitro* may not produce a lattice state resembling that in cells, either due to differences in protofilament number and/or buffer conditions and/or the lack of MAPs during polymerization. Second, the α CTT may switch between accessible and inaccessible states in both GDP- and GTP-microtubules, but the conformation switch rate differs between GDP-microtubules vs GTP-microtubules. In this scenario, Y- α CTT probes would be expected to saturate α CTT-binding sites on both GDP- and GTP-microtubules *in vitro*. Consistent with this possibility, our MD simulations show that the α CTT undergoes transient interactions with multiple sites along the microtubule lattice. Third, MAPs that bind to or induce an expanded lattice may promote Y- α CTT accessibility. Indeed, we demonstrate that the addition of expander MAPs can increase Y- α CTT accessibility both in cells (**Figure 3**) and for microtubules polymerized under *in vitro* conditions (**Figure 6E-G**).

The α CTT is sequestered through interactions with the tubulin body

We propose that the α CTT is sequestered along the microtubule through interactions with the tubulin body in a manner that precludes its recognition by the Y- α CTT probes. Indeed, our MD simulations show that the α CTT undergoes frequent interactions with the bodies of α -tubulin (*cis* interactions) and β -tubulin (*trans* interactions). These findings are consistent with an NMR study showing that the α CTT can interact with the body of the α,β -tubulin dimer (Wall et al. 2016) and with recent molecular modeling studies suggesting that the CTTs form frequent contacts with themselves and the microtubule body (Luchko et al. 2008, Freedman et al. 2011, Laurin et al. 2017). Furthermore, our data are in agreement with recent modeling work suggesting that CTT-lattice interactions are primarily observed for the α CTT interacting with the β -tubulin of the neighboring dimer towards the minus direction along a single protofilament (Sahoo and Hanson 2025). Our work extends these findings by identifying four primary sites of α CTT-tubulin interactions.

Site 1 involves interactions of the N-terminal residues of the α CTT (E441, E443) with the adjacent β -tubulin (R390, R391, K392) and our simulations indicate that these interactions are similar between GDP and GTP microtubules. Interestingly, this contact site has been previously observed by cryo-electron microscopy. Specifically, Bodey et al. determined a 9 Å structure of Taxol-stabilized bovine brain microtubules decorated with the kinesin-5 motor domain and observed density corresponding to the α CTT interacting with β -tubulin of the longitudinally neighbouring dimer (Bodey et al. 2009). More recently, Zehr and Roll-Mecak determined a 2.9 Å structure of GMPCPP-stabilized microtubules polymerized from $\alpha 1B/\beta I + \alpha 1/\beta IVb$ tubulins and observed contacts between V437, S439, and E441 of α -tubulin and residues R390 and R391 of the longitudinally adjacent β -tubulin (Zehr and Roll-Mecak 2023).

In contrast, the interactions at sites 2, 3, and 4 have not been observed in previous modeling or structural studies. These interactions involve salt bridges between acidic residues at the C-terminal end of the α CTT (E445-E450) and clusters of basic residues on the adjacent β -tubulin (site 2: K174, R380, R213, K379, and R306) or on the α -tubulin itself (site 3: K311, R308, K338, R339 and site 4: K112, R123, R156, K163, K430). We note that these interactions are energetically favorable since the salt bridges are a strong enthalpic interaction and the multitude of conformations reduces the entropic penalty. The multiplicity of interactions between the α CTT and the microtubule lattice explains why such interactions are not able to be resolved in structural studies. The transient nature of these interactions explains why the Y- α CTT probes show no difference in steady-state binding between reconstituted GMPCPP and GDP microtubules. Importantly, our simulations indicate that the *trans* interactions with site 2 and the *cis* interactions with site 4 are reduced in the GTP state, consistent with the α CTT being more accessible along a GTP expanded microtubule lattice.

Crosstalk between CTTs, MAPs, and PTMs

Overall, our data support a model in which the α CTT is generally not accessible along the GDP-microtubule but can be “exposed” by MAPs. We speculate that, in cells, the microtubule lattice is decorated by MAPs that preferentially recognize the compacted GDP-lattice state and that these compactor MAPs in turn mask the α CTT and/or stabilize α CTT interactions with the microtubule lattice. Lattice expansion, *e.g.*, through Taxol treatment or overexpression of expander MAPs, would evict compactor MAPs and result in exposure of the α CTTs. Identification of compactor MAPs that help to conceal the α CTT is a major area for future work and represents a significant challenge because our current knowledge of MAPs that bind MTs in a conformation-sensitive manner is limited. Moreover, compactor MAPs may function synergistically or in concert, complicating the identification of MAPs that regulate α CTT accessibility.

An important implication of our work is that the α CTT is not freely available to undergo post-translational modification. Indeed, we find a strong correlation between the perturbations we used to expand the microtubule lattice and the presence of Δ Y-MTs. Taxol treatment expands the microtubule lattice and promotes detyrosination (Yue et al. 2023 [\[1\]](#)) as does expression of expander MAPs but not compactor MAPs (Figure 3 [\[2\]](#)) or expression of a GTP-locked tubulin (Figure 4 [\[3\]](#)). Our findings may also apply to other PTMs of the α CTT, *i.e.*, polyglycylation and polyglutamylation. This possibility is supported by observations showing that polyglutamylation and detyrosination often co-localize in cells (Ebberink et al. 2023 [\[4\]](#)). Further pursuit of this idea is complicated by the fact that the C-terminal tails of both α - and β -tubulin can be modified with these PTMs and we generally lack tools to distinguish between α -tubulin and β -tubulin modifications. In addition, it is not known if the accessibility of the β CTT is regulated in a manner similar to the α CTT. Recent findings showing that MATCAP2/TCMP2 trims the C-terminal tail of β 1-tubulin to generate Δ 3- β 1 and that this modified β 1-tubulin localizes to specific subcellular structures strongly suggest that post-translational modification of the β CTT is also regulated (Nicot et al. 2023 [\[5\]](#)). Our work provides a conceptual roadmap to explore this interesting possibility.

A second important implication of our work is how the α CTT can spatially and temporally influence the binding of MAPs to the microtubule. For example, the molecular motors dynein and kinesin-3 KIF13B contain CAP-Gly domains and their initial microtubule interactions (on-rates) are thus facilitated by a tyrosinated α CTT (Niederstrasser et al. 2002 [\[6\]](#), McKenney et al. 2016 [\[7\]](#), Fan and McKenney 2023 [\[8\]](#)). We speculate that these motor proteins preferentially target microtubule segments where expander MAPs have exposed the α CTT. And while kinesin-1 expands the microtubule lattice (Peet et al. 2018 [\[9\]](#), Shima et al. 2018 [\[10\]](#)), and our work suggests that MAP7 also recognizes or expands the microtubule lattice and we thus speculate that the ability of MAP7 to facilitate the loading of kinesin-1 onto microtubules may involve lattice expansion in addition to protein-protein interactions between MAP7 and kinesin-1 (Tymanskyj et al. 2018 [\[11\]](#), Chaudhary et al. 2019 [\[12\]](#), Hooikaas et al. 2019 [\[13\]](#), Metivier et al. 2019 [\[14\]](#), Pan et al. 2019 [\[15\]](#), Monroy et al. 2020 [\[16\]](#)). Further work will address these and other outstanding questions about the interplay between microtubule lattice state, MAPs, and molecular motors.

Material and methods

Y- α CTT probe plasmids

The 4xCAPGly probe consists of rat CLIP1 (CLIP-170) aa 3-484 and is based on a fragment of CLIP-170 previously called H2 (Scheel et al. 1999 [\[17\]](#), Arnal et al. 2004 [\[18\]](#), Bieling et al. 2008 [\[19\]](#)). To generate a plasmid for transient mammalian expression of a C-terminally tagged 4xCAPGly probe, the DNA coding for amino acids 3-484 of rat CLIP1 (Uniprot Q9JK25) was prepared by PCR

amplification using the primers CAPGLYfor1 and CAPGLYrev1 ([Supp Table 1](#)). The PCR fragment was joined with BamHI-linearized pmEGFP-N1 or pmScarlet3-N1 vectors via Gibson Assembly (NEB) to create pN1-4×CAPGly-mEGFP and pN1-4×CAPGly-mSc3 plasmids.

To generate a stable knock-in HeLa-Kyoto cell line expressing 4xCAPGly, the DNA for 4×CAPGly-mEGFP and 4×CAPGly-mSc3 were amplified by PCR using primers CAPGLY_for2 and CAPGLY_revGFP or CAPGLY_revSc3 ([Supp Table 1](#)). The PCR fragments were inserted via Gibson Assembly (NEB) into a pEM791 vector that was digested with AgeI and BsrGI. The pEM791-4×CAPGly-mEGFP and pEM791-4×CAPGly-mSc3 plasmids were used to establish knock-in HeLa cell lines via recombination mediated cassette exchange ([Khandelia et al. 2011](#)) in order to express 4×CAPGly-mEGFP or 4×CAPGly-mSc3 proteins in a doxycycline-inducible manner.

To generate a plasmid for bacterial expression of a 6×His-tagged 4xCAPGly-mEGFP probe, the DNA coding for 4×CAPGly-mEGFP was PCR amplified from the pmEGFP-N1-4×CAPGly plasmid using primers CLIP_for3 and CLIP_rev3 ([Supp Table 1](#)). The PCR fragment was joined with a BamHI-linearized pET15b vector via Gibson Assembly (NEB).

To generate a plasmid for transient mammalian expression of the superTagRFP-tagged A1aY1 probe, a 232 bp fragment of A1aY1 lacking the start codon was synthesized (IDT) and PCR amplified using the primers A1aY1_for1 and A1aY1_rev1 ([Supp Table 1](#)). The PCR fragment was inserted into a EcoRI-linearized psuperTagRFP-C1, in which mEGFP of pmEGFP-C1 vector had been replaced with superTagRFP using Gibson Assembly, resulting in the plasmid pC1-superTagRFP-A1aY1.

To generate a stable knock-in HeLa-Kyoto cell line expressing superTagRFP-A1aY1, the coding sequence was amplified from pC1-superTagRFP-A1aY1 using the primers A1aY1_for2 and A1aY1_rev2 ([Supp Table 1](#)) and inserted into a pEM791 vector that was linearized by digestion with AgeI and BsrGI. The resulting pEM791-superTagRFP-A1aY1 plasmid was used to establish a knock-in HeLa cell line via recombination mediated cassette exchange ([Khandelia et al. 2011](#)) in order to express superTagRFP-A1aY1 protein in a doxycycline-inducible manner. To generate a plasmid for bacterial expression of the A1A1 probe, we placed the fluorophore at the C-terminus of A1aY1 based on a previously published study ([Kesarwani et al. 2020](#)). Fragments of superTagRFP and A1aY1 were separately PCR-amplified with primers A1aY1_for3 and A1aY1_rev3, and sTagRFP_for1 and sTagRFP_rev1 ([Supp Table 1](#)), respectively. These two fragments were inserted then into BamHI-linearized pET15b vector via Gibson Assembly, which yielded the construct pET15b-superTagRFP-A1aY1. This construct was used to produce the recombinant fusion protein 6xHis-A1aY1-superTagRFP.

Tubulin plasmids

The plasmid for expressing PA-tagged human TubA1A WT and E254A constructs was generated by replacing the 6xHis tag inserted between Ile42 and Gly43 in the flexible loop of TubA1A in the vectors pJM546 and pJM602 (kindly provided by Jeff Moore, University of Colorado Anschutz Medical Center) with a PA-tag (GVAMPGAEDDVV). The resulting plasmids contain PA-tagged TubA1A followed by an internal ribosome entry site (IRES) driving the expression of an EGFP reporter. The template plasmids were PCR amplified using phosphorylated primers His-PA_for and His-PA_rev ([Supp Table 1](#)), and ligated, yielding pCAGGS-internal PA-TubA1A-IRES-EGFP. For immunofluorescence studies, constructs without the IRES-GFP component were used. These versions were generated by amplifying the full vectors excluding the IRES-GFP region using phosphorylated primers IRES-GFP_removal_for and IRES-mEGFP_removal_rev ([Supp Table 1](#)), followed by ligation to re-circularize the vectors.

For GFP-tagged TubA1A expression, untagged TubA1A was amplified using primers mEGFP-TubA1A_for and mEGFP-TubA1A_rev, and inserted into a pCAGGS vector that had been linearized with primers IRES-GFP_removal_for and pCIG2_rev ([Supp Table 1](#)). The insertion was

performed via Gibson assembly, yielding the pCAGGS-mEGFP-TubA WT construct.

MAP plasmids

MAP2 and MAP7 were tagged at their N-termini with the PA tag and at their C-termini with mEGFP (PA-MAP2-mEGFP and PA-MAP7-mEGFP). The coding sequence of human MAP2 (UniProt P11137-4) was amplified from cDNA (Horizon Discovery; Clone ID 5223046) using primers MAP2_for and MAP2_rev (**Supp Table 1** [↗](#)). The coding sequence of mouse MAP7 (UniProt O88735-2) was amplified from pCAGG-MAP7(FL)-mCherry (Tymanskyj et al. 2018 [↗](#)) using primers MmMAP7_for and MmMAP7_rev for MAP7 (**Supp Table 1** [↗](#)). The PCR products were used as inserts in a Gibson assembly reaction with pPA-mEGFP-N1-EML2-S plasmid (Hotta et al. 2022 [↗](#)) that was linearized and PCR-amplified with primers PA-mEGFP-N1_for1 and PA-mEGFP-N1_rev1 (**Supp Table 1** [↗](#)).

For the Kif5C^{rigor} construct, the rat Kif5C(1-560,G235A) sequence (UniProt P56536) was amplified from pN1-KIFC(1-560,G235A) using primers Kif5C_for and Kif5C_rev (**Supp Table 1** [↗](#)). The resulting PCR product was used as an insert in a Gibson assembly with the pmEGFP-N1 vector linearized and PCR-amplified with primers PA-mEGFP-N1_for2 and PA-mEGFP-N1_rev2.

Tau, CAMSAP2, and CAMSAP3 were tagged at their N-termini with mEGFP. Human tau (UniProt P10636-6) was PCR-amplified from cDNA (Horizon Discovery, Clone ID 40007445) using primers tau_for and tau_rev (**Supp Table 1** [↗](#)). Human CAMSAP2 (UniProt Q08AD1-1) was PCR-amplified from pEGFP-C1-CAMSAP2 (Yue et al., 2023 [↗](#)) using primers CAMSAP2_for and CAMSAP2_rev (**Supp Table 1** [↗](#)). Human CAMSAP3 (UniProt Q9P1Y5-2 with N-terminal 19 amino acid deletion) was PCR-amplified from cDNA (Horizon Discovery, Clone ID, 3868695) using primers CAMSAP3_for and CAMSAP3_rev (**Supp Table 1** [↗](#)). The PCR products were used as inserts in a Gibson assembly reaction with the pPA-mEGFP-C1 vector (Hotta et al. 2022 [↗](#)) linearized by EcoRI digestion.

In some cases, transient expression of these MAPs suppressed the expression of Y-αCTT probes in the knock-in HeLa cell lines. To mitigate this, mEGFP-tagged tau, MAP2, CAMSAP2, and CAMSAP3 were further transferred into pCAGGS vectors containing a β-actin promoter. The pCAGGS vector was PCR-amplified with primers pCAGGS_for and pCAGGS_rev (**Supp Table 1** [↗](#)) and the inserts were added by Gibson assembly. The primers mEGFP_MAP_for and mEGFP_MAP_rev were used to PCR-amplify inserts of mEGFP-tau, mEGFP-CAMSAP2, and mEGFP-CAMSAP3 whereas the primers mEGFP_MAP_for and PA-MAP2-mEGFP_rev were used to PCR-amplify the insert of MAP2-mEGFP. All plasmids were validated by Sanger sequencing.

Generation of rMAb-YL1/2-EGFP antibody and Fab

We determined the protein sequences of the immunoglobulin G (IgG) heavy and light chains of the rat monoclonal anti-tyrosinated α-tubulin antibody YL1/2 using mass spectrometry (Rapid Novor). As leucine and isoleucine have the same mass, their identities were distinguished by further fragmentation at the Cβ-Cγ bond to release a propyl (leucine) or ethyl (isoleucine) group [W-ion determination, Rapid Novor (Zhokhov et al. 2017 [↗](#))]. The Clothia method was used to identify the three complementarity determining regions (CDRs) within the variable regions of each light chain and heavy chain (Chothia and Lesk 1987 [↗](#)). CDRs are hypervariable loops that form the antigen binding interface and thus determine antibody specificity and affinity. The surrounding sequences (framework regions) dictate and maintain the folded state of the variable domain such that the CDRs are properly displayed on one side of the folded protein (Zhu et al. 2025 [↗](#)).

To generate a recombinant monoclonal antibody (rMAb), the coding sequences of the HC and LC were tagged at the N-terminus with a signal sequence (MGWSCILFLVATATGVHS) for entry into the secretory pathway. The coding sequence of the LC was additionally modified by tagging the C-terminus with a GGGS linker followed by EGFP. Recombinant YL1/2-EGFP protein was produced using the CHO-Express system (Genscript). A Fab fragment of the YL1/2-EGFP protein was prepared using the Fab Preparation Kit (Thermo Fisher Scientific; Cat# 44985). Briefly,

recombinant YL1/2-EGFP protein was digested with papain for 5 hours at 37°C followed by negative purification via a Protein A column. Flow-through fractions from the Protein A column were combined and concentrated using an Amicon Ultra-10K ultrafiltration device (Millipore, Cat# UFC801024). The YL1/2^{Fab}-EGFP (dissolved in 1×PBS) was aliquoted, flash frozen, and stored at −80°C until use.

Protein Expression and Purification

The 6×His-tagged 4xCAPGly-mEGFP protein was expressed in BL21-CodonPlus-RILC *E. coli* cells (Aligent Technologies Cat# 230245). Bacterial cultures were induced with 0.5 mM IPTG at 18°C for 18 hr. Purification of the 4xCAPGly-mEGFP protein was performed based on (Bieling et al., 2008 [DOI](#)). Cell pellets were resuspended in lysis buffer [50 mM KPi (pH 7.5), 500 mM NaCl, 1 mM MgCl₂, 1 mM BME, 1 mM PMSF, 1.0 mg/ml lysozyme (Sigma-Aldrich, Cat# L6876), 1 × protease inhibitor cocktail (SIGMAFAST, Sigma-Aldrich, Cat# S8830), and Benzonase nuclease (Sigma-Aldrich, Cat# E1014)], sonicated, and clarified by centrifugation. The lysate was applied directly to Ni-NTA resin that was pre-equilibrated with lysis buffer. Once the protein was bound, the column was washed with 10 column volumes of low imidazole wash buffer [50 mM KPi (pH 7.5), 500 mM NaCl, 1 mM MgCl₂, 8.5 mM Imidazole (pH 7.4), and 1 mM BME] followed by an additional wash with 3 column volumes of high imidazole wash buffer [50 mM KPi (pH 7.5), 500 mM NaCl, 1 mM MgCl₂, 125 mM Imidazole (pH 7.4), and 1 mM BME]. Proteins were eluted with elution buffer [50 mM KPi (pH 7.5), 500 mM NaCl, 1 mM MgCl₂, 300 mM Imidazole (pH 7.4), and 1 mM BME]. Fractions containing the protein were combined and gel filtered over a HiLoad 16/600 Superdex 200 pg column (Cytiva) that was equilibrated with 50 mM KPi (pH 7.5), 150 mM NaCl, 1 mM MgCl₂, and 1 mM BME. Fractions containing the protein were pooled, concentrated with an appropriately sized MWCO centrifugal concentrator (MilliporeSigma), aliquoted, flash frozen, and stored at −80°C until use.

The 6xHis-superTagRFP-A1aY1 protein was expressed in Rosetta 2(DE3) pLysS *E. coli* cells (Novagen, Cat# 71403-3). The 6xHis-superTagRFP-A1aY1 protein was purified similarly as the 4xCAPGly proteins with the following changes: the cells were lysed in lysis buffer [50 mM KPi (pH 7.5), 500 mM NaCl, 20 mM imidazole, 1 mM PMSF, 1 × protease inhibitor cocktail (SIGMAFAST, Sigma-Aldrich), and Benzonase nuclease]. The low imidazole wash step was omitted and the column was washed with Ni-NTA wash buffer [50 mM KPi (pH 7.5), 500 mM NaCl, 20 mM imidazole, 1 mM MgCl₂, 1 mM ATP and SIGMAFAST] and protein was eluted from the column with elution buffer [50 mM KPi (pH 7.5), 500 mM NaCl, 200 mM imidazole]. The protein was separated by gel filtration using buffer [10 mM Hepes, 300 mM KCl, 1 mM DTT (pH 7.5)]. Protein concentration was determined via Bradford protein assay.

The KIF5C(1-560) protein was expressed in Sf9 cells (Thermo Fisher Scientific). The cells were cultured in suspension with serum-free sf900 II SFM medium (Thermo Fisher Scientific) supplemented with antibiotic antimycotic (Gibco) in flasks at 28°C in a non-CO₂ nonhumidified incubator with an orbital shaker platform set at 110 rpm. The KIF5C(1-560) protein consists of rat KIF5C aa 1-160 fused to Halo tag and a dual StrepII tag [KIF5C(1-560)-Halo-2xstreptII]. The construct was subcloned into the pFastBac1 vector via Gibson assembly. Baculovirus was generated according to the Bac-to-Bac system (Invitrogen). In brief, plasmids were transformed into DH10Bac *E. coli* to generate recombinant bacmids. Bacmid DNA was isolated with the HiPure Plasmid DNA miniprep kit (Invitrogen) and confirmed by PCR analysis. Recombinant bacmid DNA was transfected into Sf9 cells using Cellfectin II (Invitrogen). 7 d after transfection, the supernatant containing P1 baculovirus was collected and clarified by centrifugation at 3,000 rpm for 3 min at 4°C. The baculovirus was amplified by successive infection of Sf9 cells to generate P2 and P3 baculoviruses. Baculovirus-containing supernatants were stored at 4°C in the dark. To purify KIF5C(1-560) protein, Sf9 cells were infected with 3% P3 baculovirus (vol/vol). 3 d after infection, the cells were harvested by centrifugation for 15 min at 3,000 rpm at 4°C. The pellet was washed once with PBS and resuspended in ice-cold lysis buffer (200 mM NaCl, 4 mM MgCl₂, 0.5 mM EDTA, 1 mM EGTA, 0.5% igepal, 7% sucrose, and 20 mM imidazole-HCl, pH 7.5) supplemented with 2 mM

ATP, 1 mM PMSF, 5 mM DTT, and protease inhibitor cocktail. After 30 min incubation on ice, the lysates were clarified by ultracentrifugation for 20 min at 20,000 rpm in F12-8×50y rotor (Sorvall 3421), and the supernatants were incubated with strep-Tactin beads (Strep-Tactin XT 4Flow resin, Iba) for 1 h at 4°C with rotation. The beads were transferred to a PD-10 column and washed with wash buffer (150 mM KCl, 25 mM imidazole-HCl, pH 7.5, 5 mM MgCl₂, 1 mM EDTA, and 1 mM EGTA) supplemented with 1 mM PMSF, 3 mM DTT, 3 mM ATP, and protease inhibitor cocktail. Bound proteins were eluted in 6×0.5 mL fractions with elution buffer (25 mM KCl, 25 mM imidazole-HCl, pH 7.5, 5 mM EGTA, 2 mM MgCl₂, 2 mM DTT, 0.1 mM ATP, 1 mM PMSF, protease inhibitor cocktail and 10% glycerol) supplemented with 50 mM biotin. The fractions were separated by SDS-PAGE and fractions containing KIF5C(1-560) were combined and dialyzed in dialysis buffer (25 mM imidazole-HCl, pH 7.5, 25 mM KCl, 5 mM EGTA, 2 mM MgCl₂, 2 mM DTT, 0.1 mM ATP and 10% glycerol). After 2 hr, the buffer was changed with fresh dialysis buffer and dialyzed overnight at 4°C to remove biotin from the sample. The protein was concentrated by centrifugation, and aliquots were snap frozen in liquid nitrogen and stored in - 80°C until further use.

Western Blot Analysis

The specificity of the Y-αCTT probes for the tyrosinated αCTT was validated utilizing GST-αCTT fusion proteins in which glutathione-S-transferase (GST) was fused to human Tuba1A αCTT sequences representing tyrosinated (Y = SVEGEGEEEGEEY), detyrosinated (ΔY = SVEGEGEEEGEE), or ΔC2 (Δ2 = SVEGEGEEEGEE) tails. GST-αCTT proteins were expressed and purified from bacteria as previously described (Hotta et al. 2022 [\[1\]](#), Hotta et al. 2023 [\[2\]](#)). GST-αCTT proteins (250 ng) were separated by SDS-PAGE and transferred on to nitrocellulose membranes. The membranes were blocked with 3% skim milk in 1 × TBS supplemented with 0.1% tween-20 (TBST blocking buffer) for 1 hr at room temperature followed by incubation with primary antibodies diluted in blocking buffer for 1 hr at room temperature. Primary antibodies and dilutions: anti-tyrosinated α-tubulin rat monoclonal antibody YL1/2 (1:1000; Bio-Rad #MCA77G), recombinant monoclonal antibody rMAb-YL1/2-EGFP (2 mg/mL; Genscript), YL1/2^{Fab}-EGFP (4 mg/mL), anti-GST mouse monoclonal antibody (1:1000; Nacalai USA; 04435-26).

HeLa cell lysates were prepared by resuspending cell pellets in lysis buffer (6 mM Na₂HPO₄, 4 mM NaH₂PO₄, 2 mM EDTA, 150 mM NaCl, 1% NP40 and protease inhibitors) followed by a brief sonication and clarification via centrifugation. Cell lysates (15 μg) were separated by SDS-PAGE and transferred to nitrocellulose membranes. The membranes were blocked with 5% skim milk in PBS supplemented with 0.5% tween-20 (PBST blocking buffer), and primary antibody incubation was carried out at 4°C overnight. Primary antibodies and dilutions: anti-PA-tag rat monoclonal antibody clone NZ-1 (1:1,000; FUJIFILM Wako Pure Chemicals Cat# 016-25861), rabbit monoclonal anti-detyrosinated α-tubulin antibody clone RM444 (0.1 μg/ml; RevMAb Biosciences; Cat# 31-1335-00); mouse monoclonal anti-α-tubulin antibody clone DM1α (1:3,000; Millipore Sigma; Cat# 05-829), mouse monoclonal anti-GAPDH antibody clone G-9 (1:2,000; Santa Cruz; Cat# sc-365062), mouse monoclonal antibody anti-TagRFP clone 6A11f (1:1,000; Kerafast; Cat# EFH005), rabbit polyclonal anti-RFP (mScarlet) antibody (1:1,000; Rockland Immunochemicals; Cat# 600-401-379).

After incubation with primary antibodies, the membranes were washed 3-5 times for 5 min each with the TBST blocking buffer or PBST and then incubated with the corresponding secondary antibodies for 1 hr at room temperature. Secondary antibodies used: goat anti-mouse IgG Alexa Fluor 700 (1:5,000; Invitrogen; A-21036), goat anti-mouse IgG DyLight 800, (1:10,000; Invitrogen; SA5-10176), goat anti-rat IgG Alexa Fluor 680 (1:5,000; Invitrogen; Cat# A-21096), donkey anti-Rat IgG DyLight 800 (1:5,000; Invitrogen; SA5-10032) and goat anti-rabbit IgG IRDye 800CW (1:10,000; LICOR bio; 926-32211). For YL1/2-GFP and YL1/2-GFP^{Fab} blots, secondary antibody step was omitted. Membranes were washed 3-5 times for 5 minutes each. Fluorescence signals were detected with the Azure 600 imaging system.

Mammalian cell maintenance and transfection

COS-7 cells [male *Ceropithecus aethiops* (green monkey) kidney fibroblast, *RRID:CVCL_0224*[\[1\]](#)] were grown in Dulbecco's modified Eagle medium (DMEM, Gibco 11960-044) supplemented with 10% (vol/vol) Fetal Clone III (HyClone SH3010903) and 2 mM GlutaMAX (L-alanyl-L-glutamine dipeptide in 0.85% NaCl, Gibco 35050061).

HeLa Kyoto cells [female *Homo sapiens* (*RRID:CVCL_1922*[\[2\]](#))] were maintained in DMEM (Gibco; Cat# 11965118) containing 10% fetal bovine serum (FBS) (Cytiva; SH3007103T), and 1% Penicillin-Streptomycin (Gibco, Cat# 15140122).

HeLa S3 cells [female *Homo sapiens* (*RRID:CVCL_0058*[\[3\]](#)) ATCC (CCL-2.2)] were grown in suspension in DMEM high glucose (Invitrogen 11965092) supplemented with 10% (vol/vol) Fetal Clone III Serum (HyClone SH3010903), GlutaMAX (Invitrogen 35050061), and 1% Penicillin/Streptomycin (Invitrogen 15140163).

HeLa knock-in cell lines expressing 4×CAPGly-mEGFP, 4×CAPGly-mSc3 and superTagRFP-A1aY1 in a doxycycline-inducible manner were maintained in DMEM (Gibco; Cat# 11965118) containing 10% FBS (Cytiva; SH3007103T), and 1% Penicillin-Streptomycin (Gibco, Cat# 15140122) and 1 mg/mL puromycin (Sigma-Aldrich; Cat# P8833). The expression of each transgene in each knock-in cell line was induced via the addition of 2 µg/mL doxycycline (Thermo Fisher Scientific; Cat# BP26531).

All cell lines were maintained in the presence of 5% CO₂ at 37°C. All cell lines were screened and found negative for mycoplasma contamination. HeLa and COS-7 cells were transfected with Lipofectamine 2000 (Thermo Fisher Scientific; Cat# 11668019), according to the manufacturer's instructions. Briefly, 0.5–1 µg of plasmid DNA was diluted in 250 µL of Opti-MEM (Gibco; 31985070). Following a 5 min incubation at room temperature, 250 µL of Opti-MEM containing 5 µL of Lipofectamine 2000 was added to the plasmid DNA and incubated for 20 min at room temperature. The entire reaction mixture was added directly to the dish of cells containing 1 mL of Opti-MEM and incubated for 3.5 hours at 37°C before exchanging the media to DMEM supplemented with 10% FBS and 1% Penicillin-Streptomycin.

Total internal reflection fluorescence microscopy

All assays used HeLa microtubules. For this, tubulin was purified from HeLa-S3 cells using a GST-TOG column and polymerized into microtubules as described (Thomas et al., 2025[\[4\]](#)). A flow cell (~10 µL volume) was assembled by attaching a clean #1.5 coverslip (Fisher Scientific) to a glass slide (Fisher Scientific) with two strips of double-sided tape. All imaging was performed at room temperature. *In vitro* assays were performed on an inverted Nikon Ti-E/B total internal reflection fluorescence microscope with a perfect focus system, a 100 × 1.49 NA oil immersion TIRF objective, three 20 mW diode lasers (488 nm, 561 nm, and 640 nm) and EMCCD camera (iXon+ DU879; Andor). Image acquisition was controlled using Nikon Elements software.

To validate the sensitivity of the Y-αCTT probes to the tyrosinated vs detyrosinated state of α-tubulin, AlexaFluor 488- or 647-labeled microtubules were generated. Taxol-stabilized microtubules were incubated in a flow chamber with cell lysate overexpressing VASH1 and SVBP for 15 minutes to generate ΔY-microtubules (Yue et al. 2023[\[5\]](#), Thomas et al. 2025[\[4\]](#)). The chambers were washed with 0.1 mg/ml casein and 10 µM paclitaxel in BRB80 to remove VASH1/SVBP enzyme. Then an imaging mixture containing YL1/2^{Fab}-EGFP (66 nM) or 4xCAPGly-mEGFP (15 nM) proteins in 10 µM paclitaxel in BRB80 supplemented with 0.1 mg/ml casein and oxygen scavengers (1 mM DTT, 1 mM MgCl₂, 10 mM glucose, 0.2 mg/ml glucose oxidase, and 0.08 mg/ml catalase) was flowed into the chambers and imaged.

To compare Y- α CTT probe binding to GMPCPP vs GDP microtubules, microtubules were polymerized from HeLa tubulin in the presence of GMPCPP or GDP and in the presence of AlexaFluor 568-labeled HeLa tubulin or AlexaFluor 647-labeled HeLa tubulin (Thomas et al., 2025). The chambers were prepared by first flowing in a recombinant anti-TubB3 Tuj1 antibody (gift from Jeffrey Moore, University of Colorado Anschutz Medical Center) diluted 1:50 in BRB80 and incubating for 5-10 min. Chambers were washed with 25% glycerol in BRB80, and then a mixture of GMPCPP and GDP microtubules was added and incubated for 5-10 min. The chambers were washed with 25% glycerol/BRB80 and then YL1/2^{Fab}-EGFP (final concentration of 66 nM) or 4xCAPGly-mEGFP (final concentration of 15 nM) proteins were added in 25% glycerol/BRB80 supplemented with 0.1 mg/ml casein and oxygen scavengers (1 mM DTT, 1 mM MgCl₂, 10 mM glucose, 0.2 mg/ml glucose oxidase, and 0.08 mg/ml catalase).

To examine whether the binding behavior of YL1/2^{Fab}-EGFP on glycerol-stabilized GDP-MTs is regulated by KIF5C(1-560)-Halo protein, GDP-MTs were assembled from HeLa tubulin and stabilized in 25% glycerol/BRB80. The microtubules were introduced into a flow cell and incubated for 3 min at room temperature to allow for nonspecific adsorption to the coverslips. After washing with blocking buffer [1 mg/mL casein and 25% glycerol in P12 buffer (12 mM PIPES/KOH pH 6.8, 2 mM MgCl₂, 1 mM EGTA)], the flow cell was infused with imaging buffer (P12 buffer supplemented with 25% glycerol, 0.3 mg/ml casein, and oxygen scavenger mix) with YL1/2^{Fab}-EGFP (4 μ g/mL) under four conditions: 1) no KIF5C, 2) weakly-bound KIF5C [100 nM KIF5C, 2 mM ADP (Sigma, A2754), 2 units/mL hexokinase (Sigma, H5000)] 3) strongly-bound KIF5C [100 nM KIF5C, 6 units/mL apyrase (Sigma, A6535)] or 4) stepping KIF5C [100 nM KIF5C, 2 mM ATP (Sigma, A7699)]. After 3 min, the flow cell was then sealed with molten paraffin wax and imaged. For the wash out experiments, the flow cell was infused with imaging buffer containing weakly-bound KIF5C [100 nM KIF5C, 2 mM ADP (Sigma, A2754), 2 units/mL hexokinase (Sigma, H5000)] or strongly-bound KIF5C [100 nM KIF5C, 6 units/mL apyrase (Sigma, A6535)]. After 3 min, the flow cell was washed with blocking buffer supplemented with 300 mM KCl, followed by imaging buffer supplemented with 52 nM YL1/2^{Fab}. The flow cell was then sealed with molten paraffin wax and imaged. Fluorescence intensities along the microtubules were measured using Fiji/ImageJ (width = 3 pixels), and the fluorescence intensity of an adjacent region was subtracted to account for background noise.

Live-cell imaging of tail and lattice sensors

For live cell imaging, cells were plated onto poly-D-lysine coated glass-bottom dishes (MatTek #P35GC-1.5-14-C) 24-48 hr before imaging and transfected with plasmids 18-24 hr prior to imaging. HeLa stable cell lines were induced for 15 hr with 2 μ g/mL doxycycline prior to imaging to express superTagRFP-A1aY1 or 4xCAPGly-mEGFP proteins. In preparation for imaging, each dish was washed with 1 mL of warm Leibovitz's L-15 Medium (Gibco; Cat# 21083027) supplemented with 10% FBS (Cytiva; SH3007103T), and 1% Penicillin-Streptomycin (Gibco; Cat# 15140122). Cells were then incubated in imaging medium for 10 min prior to imaging.

To assess changes in Y- α CTT probe localization, cells exhibiting similar levels of fluorescence were identified prior to imaging. Single optical sections were imaged. The localization of each Y- α CTT probe was monitored before treatment and upon treatment with 10 mM Taxol or 0.3% DMSO (vehicle control). After 15 min, the same optical sections of each cell were imaged to directly compare the localization and intensity of each probe along the microtubule lattice in a compacted (before) and expanded (after) state. All cells were imaged and maintained at 37°C using a DeltaVision Elite microscope system equipped with an Olympus Plan Apo N 60 $\times$ 1.42 NA oil immersion objective.

For live imaging of COS-7 cells, cells were plated onto glass-bottom dishes (MatTek #P35G-1.5-14-C) 48 hr prior to imaging. Cells were transfected with 0.5 μ g of EGFP-N1 and 0.2 μ g of Y- α CTT plasmid 24 hr prior to imaging. Cells were transferred to warm Leibovitz's L-15 medium for imaging.

Images were acquired on an inverted epifluorescence microscope (Nikon TE200E) with a 60x, 1.4 NA oil-immersion objective, TokaiHit stage-top incubator (INUG2A-GILCS) set to 37°C, and an Orca Flash4 OLT digital CMOS camera (Hamamatsu).

Cell fixation and immunofluorescence

COS-7 cells plated onto glass coverslips were rinsed with PBS⁺ (PBS with 0.9 mM CaCl₂, 0.5 mM MgCl₂), then fixed and permeabilized simultaneously in pre-chilled methanol (MeOH) for 10 min at -20°C. Cells were then either washed twice with PBS⁺, rinsed with ddH₂O, dabbed on a kimwipe to remove excess H₂O, and mounted onto a slide with Prolong Gold (Invitrogen P36930) or washed twice with PBS⁺, blocked briefly with 0.2% fish skin gelatin (FSG, Sigma G7765) in PBS⁺ and subjected to immunostaining. Note that soluble EGFP used as a control in these experiments is not preserved with MeOH fixation. Mouse anti-β-tubulin (DSHB #E7, 1:1,000) primary antibody was diluted in 0.2%FSG in PBS⁺ and applied to cells for 1 hr. Cells were washed three times with 0.2% FSG in PBS⁺ and then subjected to secondary antibody staining for 1 hr. Alexa Fluor 680-conjugated AffiniPure Donkey Anti-Mouse IgG (H+L) (Jackson Immuno Research #715-625-150) was diluted 1:500 in 0.2% FSG in PBS⁺. Finally, cells were washed an additional three times with 0.2% FSG in PBS⁺ and then twice with PBS⁺. Coverslips were dipped in ddH₂O, dabbed on a kimwipe to remove excess water, and mounted with Prolong Gold.

HeLa cells were cultured on coverslips 24 hr prior to transfection. Approximately 20 hr post-transfection, cells were fixed with methanol for 10 min at -20°C. Following fixation, cells were rehydrated in TBS supplemented with 0.1% Triton X-100 (TBS-Triton), and then blocked for 15 min with 2% BSA in TBS-Triton. All primary and secondary antibodies were diluted in 2% BSA in TBS-Triton. Detyrosinated microtubules were stained with a recombinant rabbit monoclonal anti-detyrosinated α-tubulin antibody (RevMab Biosciences, Cat# RM444; 0.5 μg/ml) for 1 hr followed by goat anti-rabbit IgG Alexa Fluor 594 (Invitrogen, Cat# A-11012; 1:1,000) for 30 min. Overexpressed MAPs were stained with a chicken polyclonal anti-GFP antibody (Aves labs, Cat# GFP-1010; 1:1,000) for 1 hr, followed by goat anti-chicken IgY Alexa Fluor 488 (Invitrogen, Cat# A-11039; 1:1,000) for 30 min. PA-tagged TubA was stained with a rat monoclonal anti-PA-tag antibody, clone NZ-1 (FUJIFILM Wako Pure Chemical, Cat# 016-25861; 1:500) for 1 hr and goat anti-rat IgG Alexa Fluor 488 (Invitrogen, Cat# A-11006; 1:1,000) for 30 min. Finally, total microtubules were stained with mouse monoclonal anti-α-tubulin antibody, clone DM1 α conjugated with Alexa Fluor 647 (Millipore Sigma, Cat# 05-829-AF647; 1:500) for 30 min. DNA was counterstained with Hoechst, and cells were mounted with Prolong Diamond. Images were obtained with a DeltaVision microscope equipped with an Olympus Plan Apo N 60x/1.42 oil immersion lens. Images were subsequently deconvolved, and single optical sections are presented.

Quantification of αCTT probes and detyrosination in live cells

Microtubule-bound A1aY1 or CAPGly sensors in Taxol-treated or TubA-expressing cells were quantified as the total length of microtubule regions labeled by Y-αCTT probe per unit cell area. Except for sTagRFP-A1aY1 in the Taxol treatment (see below), we measured microtubule density using an image processing method developed recently (Horiuchi et al. 2025 [\[1\]](#)). First, to enhance the visualized microtubule signal, raw microscopic images were subjected to 2D segmentation using a deep learning-based function in the image analysis software AIVIA (DRVision, Bellevue, WA, USA). This deep learning model, based on RCA-UNet for image transformation, was trained on a dataset consisting of raw images acquired using various microtubule-targeting probes and their corresponding binarized images, in which visualized microtubule regions were manually segmented (Horiuchi et al. 2025 [\[1\]](#)). The enhanced images were then binarized using Otsu's thresholding in ImageJ and skeletonized using the ImageJ plug-in Lpx_bilevelThin (Higaki et al. 2010 [\[2\]](#)). Cell regions were manually segmented using the Polygon Selection tool in ImageJ. Finally, microtubule density (referred to as occupancy) was calculated as the ratio of the skeletonized microtubule length to the total cell area.

Microtubule-bound superTagRFP-A1aY1 sensor in the Taxol treatment experiment was quantified using Fiji Software (Schindelin et al. 2012 [DOI](#)). All images were sequentially processed using a 5-pixel rolling ball radius background subtraction, 2.0-pixel Gaussian blur, and brightness and contrast set to a minimum and maximum of 50 and 990, respectively. Images were then converted to 8-bit grayscale and microtubule-bound sensor was detected using the Fiji Ridge Detection plugin (Wagner et al. 2017 [DOI](#)) with the following settings: line width of 2.0, high contrast of 100.0, low contrast of 10.0, sigma of 1.08, lower threshold of 0.68, upper threshold of 6.97, minimum line length of 10.0, and slope method of overlap resolution. The images were binarized and total signal (IntDen) was recorded for each image and normalized to cell area.

We quantified colocalization between MAPs and Y-aCTT probes (**Figure 3** [DOI](#)), or MAPs and detyrosination (**Figure 4** [DOI](#)) using the threshold overlap score (TOS). This metric was calculated with the imageJ plugin EzColocalization (Stauffer et al. 2018 [DOI](#)). For analysis, TOS was determined based on the top 10th percentile of signal per cell, with cell boundaries defined manually.

To quantify microtubule detyrosination in cells expressing MAPs, we measured detyrosination signal exclusively on MAP-decorated microtubules, rather than on the entire microtubule network within the cells. Our rationale for this approach is that some MAPs exhibit localization only on specific subsets of microtubules. Therefore, measuring detyrosination against total cellular microtubules, as we performed previously (Yue et al. 2023 [DOI](#)), would inaccurately represent the net change in detyrosination directly influenced by MAP localization. To achieve this targeted measurement, we first extracted MAP-localized microtubule pixels by applying a consistent threshold across all MAP images, creating specific masks. These masks were then applied to both the total microtubule channel (visualized by anti- α -tubulin antibody DM1 α) and the detyrosinated microtubule channel to obtain mean intensities for each cell.

Finally, the relative intensities of detyrosination signals were plotted after normalization against the corresponding total microtubule intensities within the masked regions. In contrast, for the E254A TubA experiments, both wild-type and E254A PA-tagged TubA constructs showed broad incorporation into microtubules throughout the entire cell. Consequently, mean detyrosination was measured on a per-cell basis, as described previously (Yue et al. 2023 [DOI](#)).

Molecular Dynamics simulations

The starting point for our GDP simulations was the structure of the GDP microtubule [PDB: 7SJ7 (LaFrance et al. 2022 [DOI](#))]. To create the GTP-microtubule, we used the GMPCPP-structure [PDB: 6DPU (Zhang et al. 2018 [DOI](#))] as a starting point, converted the GMPCPP to GTP, and fit the structure to the 7SJ7 microtubule so that it had the same lattice structure. C-terminal tails were added to every tubulin subunit. For all systems, rigid-body fitting was used to make a complete 13-protofilament, 3-start microtubule ring. This single ring was then shifted by the dimer repeat distance in order to make a 3-ring, 39-dimer microtubule fragment. For the GDP-microtubules, the 3-ring structure was converted to an “infinite” microtubule (Wells and Aksimentiev 2010 [DOI](#), Igaev and Grubmuller 2020 [DOI](#)) by putting it in a periodic box where the axial length was three times the dimer repeat distance. Due to the inherent twist of the GTP-microtubule, we could not create an “infinite” microtubule since the protofilaments would not match up with their image. Instead, the 39-dimer GTP system was solvated in a box with 20 Å padding on each side.

All systems were solvated using TIP3P water and ionized with Na⁺ and Cl⁻ in order to both neutralize the system and set the ionic strength to 50 mM. Simulations were carried out using NAMD (Phillips et al. 2020 [DOI](#)) using the CHARMM36 (Best et al. 2012 [DOI](#)) force field. Following minimization and heating, we performed a short 10 ns equilibrium in an NpT ensemble with 1 atm pressure at 300 K. We then ran two independent trajectories for 260 ns and 330 ns, respectively, in the GDP-microtubule system, and one 360 ns trajectory in the GTP-microtubule

system, utilizing hydrogen mass repartitioning (Hopkins et al. 2015 [↗](#)) to allow for 4 fs time steps. All analysis was done using bio3D (Grant et al. 2006 [↗](#)) and R (<https://www.r-project.org/> [↗](#)). Images and movies were created using VMD (Humphrey et al. 1996 [↗](#)).

Data analysis, statistics, and presentation

All data were plotted and statistical tests were performed using GraphPad Prism (version 10.4.1; GraphPad Software). The statistical tests, sample size, and number of replicates used for each experiment are described in each figure legend. Figures were made in Adobe Illustrator 2025 (version 29.0; Adobe).

Data availability

All data are available upon request

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Additional information

Contributor Roles

TH Investigation; Validation; Formal analysis; Visualisation; Writing – review & editing;

ECT Investigation; Validation; Formal analysis; Visualisation; Writing – review & editing;

MLP Investigation; Validation; Formal analysis; Visualisation; Writing – review & editing;

YY Investigation; Validation; Formal analysis; Visualisation; Writing – review & editing;

PD Investigation; Validation; Formal analysis; Visualisation; Writing – review & editing;

TLB Investigation; Validation; Formal analysis; Visualisation; Writing – review & editing;

MC Conceptualization; Writing – review & editing;

MD Conceptualization; Writing – review & editing;

RH Methodology; Formal analysis;

TH Methodology; Formal analysis;

DS Conceptualization; Supervision; Funding acquisition; Writing – review & editing;

RO Conceptualization; Supervision; Project administration; Funding acquisition; Writing – original draft preparation; Writing – review & editing;

KJV Conceptualization; Supervision; Project administration; Funding acquisition; Writing – original draft preparation; Writing – review & editing;

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MD:

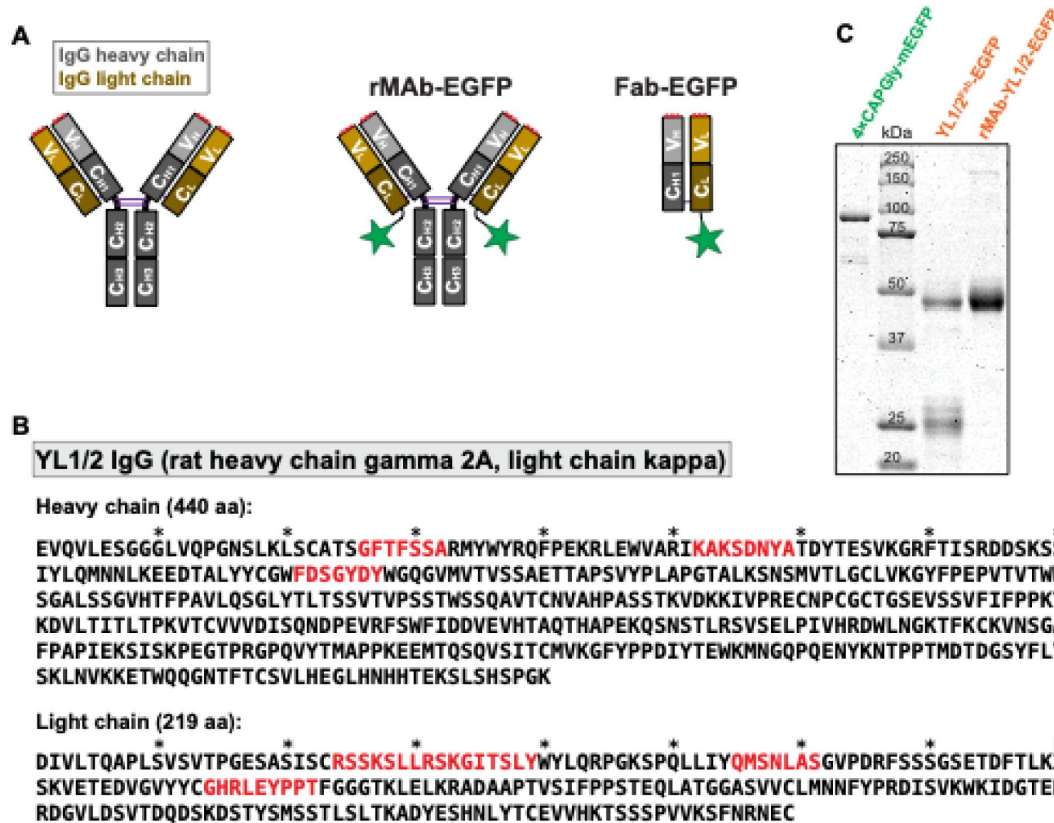
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The funding sources were not involved in study design, data collection and interpretation, or the decision to submit the work for publication.

Additional files

Supp Movie 1. *Movie from MD simulations. Representative movie from MD simulations showing interactions of the α CTT with the microtubule body over 240 ns. The tubulin body is shown in cartoon and colored gray. The α CTT is shown in stick and colored yellow with the glutamate sidechains in red. The residues in site 1 (green), site 2 (cyan), and site 3 (magenta) appear as spheres when the α CTT is forming salt bridges with those residues.* [🔗](#)

Supplementary figures and table

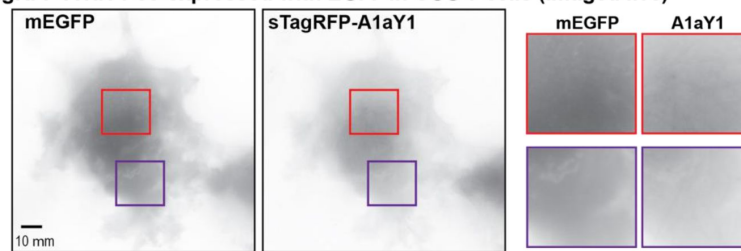


Supp Figure 1.

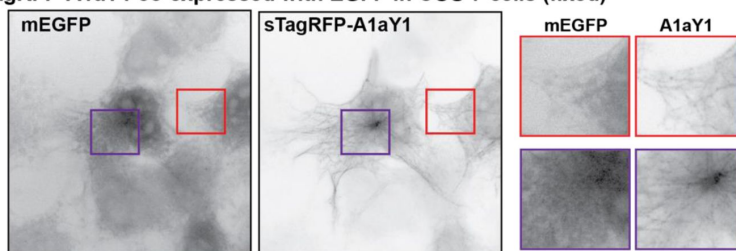
Recombinant YL1/2 antibody and purified probes.

(A) Schematic of a typical mammalian IgG molecule containing two heavy (H) and two light (L) chains. Light chains are comprised of one variable (V_L , light orange) and one constant (C_L , dark orange) region. Heavy chains are comprised of one variable (V_H , light gray) and three constant (C_{H1-3} , dark gray) regions. Red dots: complementarity determining regions (CDRs). Purple lines: disulfide bonds. rMab-EGFP: recombinant monoclonal antibody (rMab) with EGFP fused to the C-terminus of the light chain. Fab-EGFP: Fragment antibody binding (Fab) produced from rMab-EGFP by papain cleavage. (B) Experimentally-determined YL1/2 protein sequence. The deduced amino acid sequences of YL1/2 IgG heavy and light chains are shown. The red text indicates the CDRs involved in antigen recognition. Asterisks demarcate every 10 aa. (C) Coomassie-stained SDS-PAGE gel of purified proteins.

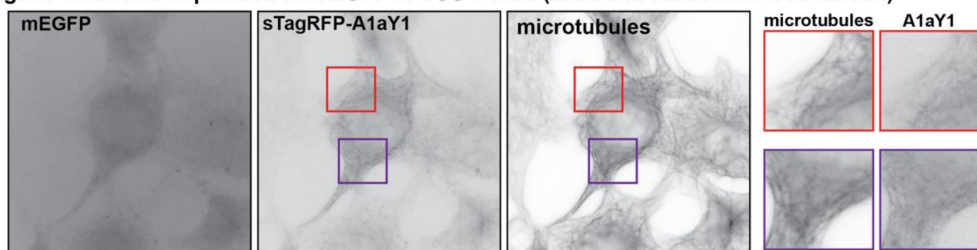
A. sTagRFP-A1aY1 co-expressed with EGFP in COS-7 cells (imaged live)



B. sTagRFP-A1aY1 co-expressed with EGFP in COS-7 cells (fixed)



C. sTagRFP-A1aY1 co-expressed with EGFP in COS-7 cells (fixed and stained for total tubulin)

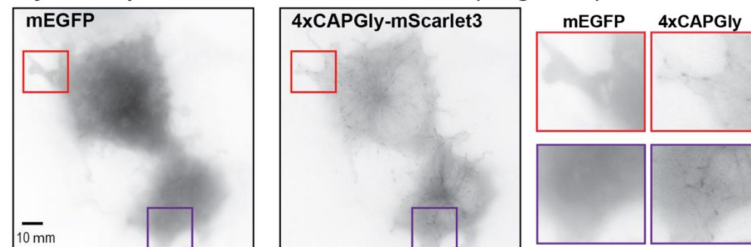


Supp Figure 2.

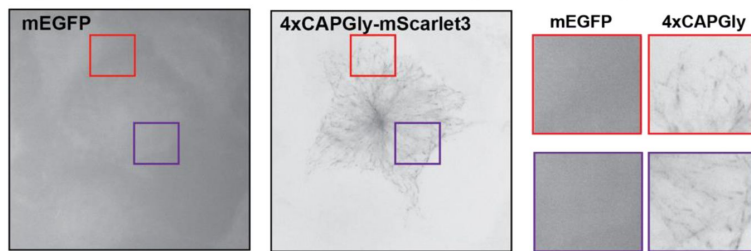
A1aY1 sensor must be imaged in live cells.

COS-7 cells expressing sTagRFP-A1aY1 and mEGFP were (A) imaged live, (B) fixed and mounted, or (C) fixed and stained for total tubulin (microtubules). Scale bar: 10 μm. Magnified views of the red and purple boxed regions are shown to the right.

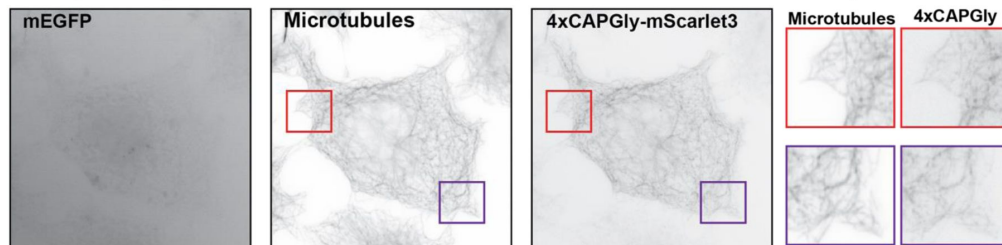
A. 4xCAPGly-mSc expressed with EGFP in COS-7 cells (imaged live)



B. 4xCAPGly-mSc expressed with EGFP in COS-7 cells (fix and mount)



C. 4xCAPGly-mSc expressed with EGFP in COS-7 cells (fixed and stained for total tubulin)



Supp Figure 3.

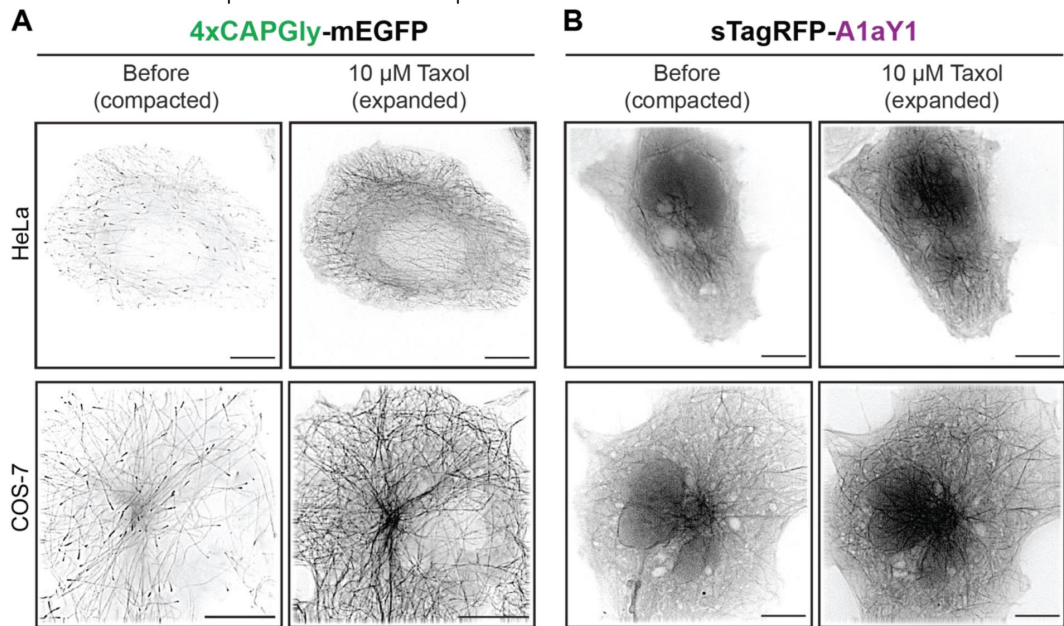
4xCAPGly sensor must be imaged in live cells.

COS-7 cells expressing 4xCAPGly-mSc3 and mEGFP were (A) imaged live, (B) fixed and mounted, or (C) fixed and stained for total tubulin (microtubules). Scale bar: 10 μm. Magnified views of the red and purple boxed regions are shown to the right.

Supp Figure 4.

Transiently-expressed probes bind to the microtubule lattice after Taxol expansion.

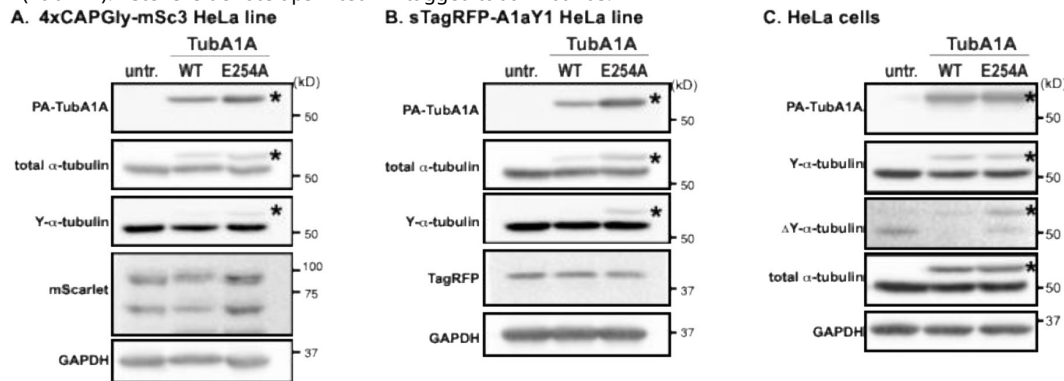
Representative images of (A) 4xCAPGly-mEGFP or (B) sTagRFP-A1aY1 probes imaged live in (top) HeLa or (bottom) COS-7 cells before or after addition of 10 μ M Taxol. Scale bars: 10 μ m.

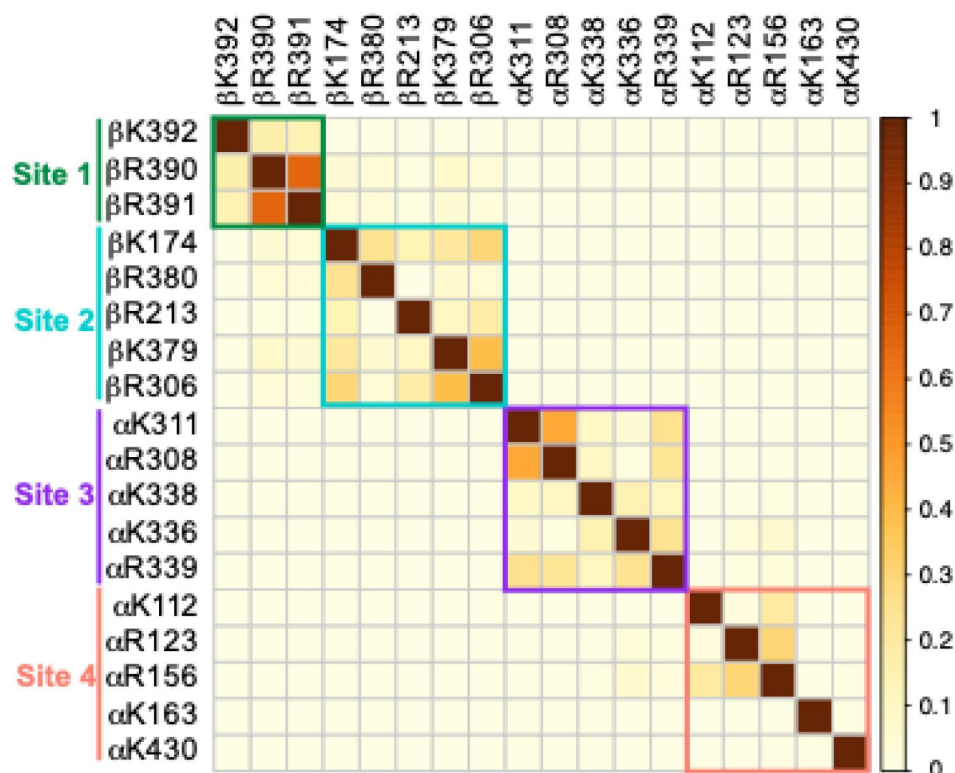


Supp Figure 5.

Western blot of HeLa cells expressing internal PA-tagged tubulin.

Whole cell lysates were prepared from (A) 4xCAPGly-mSc3 stable HeLa cells, (B) sTagRFP-A1aY1 stable HeLa cells, or (C) HeLa cells. In all cases, the cells were untransfected (untr.) or transfected with plasmids for expressing PA-tagged WT or E254A α -tubulin (TubA1A). Asterisks denote upshifted PA-tagged tubulin bands.





Supp Figure 6

The Y-αCTT primarily contacts four sites along a GDP microtubule lattice.

Jaccard index plot indicating the frequency of two residues simultaneously forming salt bridges with the Y-αCTT based on MD simulations of a GDP microtubule lattice. The scale represents the Jaccard index where an index of 0 indicates that the two residues are never interacting with the Y-αCTT at the same time and an index of 1 indicates that when one of the two residues is interacting with the Y-αCTT, the other is also interacting with the Y-αCTT.

CAPGLY_for1	CTGCAGTCGACGGTACCGCGGGCCCGGATGCTGAAACCCAGCGGGCTG
CAPGLY_rev1	CACCTCCCGATCCACCACCGCCCTCTGGAGTTTGTCAGCTTTGGTCTTT TC
CAPGLY_for2	GTACCGCGGGCCCGGGATCCAATG CTGAAACCCAGCGGGCTG
CAPGLY_revGFP	GCCTGCACCTGAGTGTTTACTTTTACTTTGTACAGCTC GTCCATGC
CAPGLY_revSc3	CTGCACCTGAGTGTTTACTTTTAGGAGCCACCGGAGCC
CAPGLY_for3	GCGGCAGCCA TATGCTCGAGCTGAAACCCAGCGGGCTGAAG
CAPGLY_rev3	CTTTCGGGCTTTGTTAGCAGCCG TTAAGTTGTACAGCTCGTCCATGC
A1aY1_for1	GATCTCGAGCTCAAGCTTCGGCAACAGTCAAGTTCAAATAC
A1aY1_rev1	CGCGGTACCGTCGACTGCAGTTACTTCTTCTGCTTCTCC
A1aY1_for2	TACCGCGGGCCCGGGATCCAATGGTGAGCAAGGGCGAG
A1aY1_rev2	CTGCACCTGAGTGTTTACTTTTACTTCTTCTGCTTCTCCAGC
A1aY1_for3	GCGGCAGCCATATGCTCGAGATGGCAACAGTCAAGTTCAAATAC
A1aY1_rev3	CCATGGTACCCTTCTTCTGCTTCTCCAG
sTagRFP_for1	GCAGAAGAAGGGTACCATGGTGAGCAAGGGCGAG
sTagRFP_rev1	TCGGGCTTTGTTAGCAGCCGTTACTTGTACAGCTCGTCCATG
His-PA_for	TGCCGAAGATGATGTGGTGGGGGAGGAGATGATTCCTTC
His-PA_rev	CCTGGCATGGCAACGCCAATGGTCTTGTCACTTGGCATC
IRES-GFP_removal_for	AGCGGCCGCAATTCACCTCCTC
IRES-mEGFP_removal_rev	TTAGTATTCCTCTCCTTCTTC
mEGFP-TubA1A_for	CTCAAGCTTCGAATTCTGCAATGGTGAGCAAGGGCGAG
mEGFP-TubA1A_rev	AGGAGTGAATTGCGGCCGCTTTAGTATTCCTCTCCTTCTTCCTCAC
pCIG2_rev	TGCAGAATTCTGAAGCTTGAGC
PA-mEGFP-N1_for1	GGTGGAGGCGGTTTCAGGC
PA-mEGFP-N1_rev1	CACCACATCATCTTCGGCACC
MAP2_for	GTGCCGAAGATGATGTGGTGATGGCAGATGAACGGAAAG
MAP2_rev	CCGCCTGAACCGCCTCCACCCAAGCCCTGCTTAGCGAG
MmMAP7_for	GTGCCGAAGATGATGTGGTGGCGGAGCAGGGAGCTGGC
MmMAP7_rev	CCGCCTGAACCGCCTCCACCTATAACTTCTGCGGTCTGTTGTGTCTGCAC
PA-mEGFP-N1_for2	GGCGTTGCCATGCCAGGTGCCGAAGATGATGTGGTGAGCAAGGGCGGAGG AGCTGTTT
PA-mEGFP-N1_rev2	CATGGTGGCGACCGGTGG
Kif5C_for	ATCCACCGGTGCGCCACCATGGCGGATCCAGCCGAATGC
Kif5C_rev	GCACCTGGCATGGCAACGCCCTCGAGCGGATCCCGGGC
tau_for	GATCTCGAGCTCAAGCTTCGATGGCTGAGCCCCGCCAG
tau_rev	CGCGGTACCGTCGACTGCAGTCACAAACCCTGCTTGCCAG
CAMSAP2_for	GATCTCGAGCTCAAGCTTCGATGGGGGATGCTGCAGACCCCA
CAMSAP2_rev	CGCGGTACCGTCGACTGCAGCTATGCCTTAGTGGGTAAAAGTTTTTGG
CAMSAP3_for	GATCTCGAGCTCAAGCTTCGATGGAGATCAAGTCGCTGGACCA GTACGATTTCTCGCGG
CAMSAP3_rev	CGCGGTACCGTCGACTGCAGCTATTTGGGAGTGCCGCC
pCAGGS_for	AGCGGCCGCAATTCACCTCCTC
pCAGGS_rev	AATTCGAAGCTTGAGCTCGAG
PA-mEGFP_MAP_for	TCGAGCTCAAGCTTCGAATTATGGCGTTGCCATGCCA
PA-mEGFP_MAP_rev	AGGAGTGAATTGCGGCCGCTTTACTTGTACAGCTCGTCCATGC
PA-MAP2-mEGFP_rev	AGGAGTGAATTGCGGCCGCTTTACTTGTACAGCTCGTCCATGC

Supplementary Table 1.

Primers used in cloning.

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Reviewer #1 (Public review):

Summary:

This is a careful and comprehensive study demonstrating that effector-dependent conformational switching of the MT lattice from compacted to expanded deploys the alpha tubulin C-terminal tails so as to enhance their ability to bind interactors.

Strengths:

The authors use 3 different sensors for the exposure of the alpha CTTs. They show that all 3 sensors report exposure of the alpha CTTs when the lattice is expanded by GMPCPP, or KIF1C, or a hydrolysis-deficient tubulin. They demonstrate that expansion-dependent exposure of the alpha CTTs works in tissue culture cells as well as in vitro.

Weaknesses:

There is no information on the status of the beta tubulin CTTs. The study is done with mixed isotype microtubules, both in cells and in vitro. It remains unclear whether all the alpha tubulins in a mixed isotype microtubule lattice behave equivalently, or whether the effect is tubulin isotype-dependent. It remains unclear whether local binding of effectors can locally expand the lattice and locally expose the alpha CTTs.

Appraisal:

The authors have gone to considerable lengths to test their hypothesis that microtubule expansion favours deployment of the alpha tubulin C-terminal tail, allowing its interactors, including detyrosinase enzymes, to bind. There is a real prospect that this will change thinking in the field. One very interesting possibility, touched on by the authors, is that the requirement for MAP7 to engage kinesin with the MT might include a direct effect of MAP7 on lattice expansion.

Impact:

The possibility that the interactions of MAPs and motors with a particular MT or region feed forward to determine its future interaction patterns is made much more real. Genuinely exciting.

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Reviewer #2 (Public review):

The unstructured α - and β -tubulin C-terminal tails (CTTs), which differ between tubulin isoforms, extend from the surface of the microtubule, are post-translationally modified, and help regulate the function of MAPs and motors. Their dynamics and extent of interactions with the microtubule lattice are not well understood. Hotta et al. explore this using a set of three distinct probes that bind to the CTTs of tyrosinated (native) α -tubulin. Under normal cellular conditions, these probes associate with microtubules only to a limited extent, but this binding can be enhanced by various manipulations thought to alter the tubulin lattice conformation (expanded or compact). These include small-molecule treatment (Taxol), changes in nucleotide state, and the binding of microtubule-associated proteins and motors. Overall, the authors conclude that microtubule lattice "expanders" promote probe binding, suggesting that the CTT is generally more accessible under these conditions. Consistent with

this, dephosphorylation is enhanced. Mechanistically, molecular dynamics simulations indicate that the CTT may interact with the microtubule lattice at several sites, and that these interactions are affected by the tubulin nucleotide state.

Strengths:

Key strengths of the work include the use of three distinct probes that yield broadly consistent findings, and a wide variety of experimental manipulations (drugs, motors, MAPs) that collectively support the authors' conclusions, alongside a careful quantitative approach.

Weaknesses:

The challenges of studying the dynamics of a short, intrinsically disordered protein region within the complex environment of the cellular microtubule lattice, amid numerous other binders and regulators, should not be understated. While it is very plausible that the probes report on CTT accessibility as proposed, the possibility of confounding factors (e.g., effects on MAP or motor binding) cannot be ruled out. Sensitivity to the expression level clearly introduces additional complications. Likewise, for each individual "expander" or "compactor" manipulation, one must consider indirect consequences (e.g., masking of binding sites) in addition to direct effects on the lattice; however, this risk is mitigated by the collective observations all pointing in the same direction.

The discussion does a good job of placing the findings in context and acknowledging relevant caveats and limitations. Overall, this study introduces an interesting and provocative concept, well supported by experimental data, and provides a strong foundation for future work. This will be a valuable contribution to the field.

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Reviewer #3 (Public review):

Summary:

In this study, the authors investigate how the structural state of the microtubule lattice influences the accessibility of the α -tubulin C-terminal tail (CTT). By developing and applying new biosensors, they reveal that the tyrosinated CTT is largely inaccessible under normal conditions but becomes more accessible upon changes to the tubulin conformational state induced by taxol treatment, MAP expression, or GTP-hydrolysis-deficient tubulin. The combination of live imaging, biochemical assays, and simulations suggests that the lattice conformation regulates the exposure of the CTT, providing a potential mechanism for modulating interactions with microtubule-associated proteins. The work addresses a highly topical question in the microtubule field and proposes a new conceptual link between lattice spacing and tail accessibility for tubulin post-translational modification.

Strengths:

- (1) The study targets a highly relevant and emerging topic-the structural plasticity of the microtubule lattice and its regulatory implications.
- (2) The biosensor design represents a methodological advance, enabling direct visualization of CTT accessibility in living cells.
- (3) Integration of imaging, biochemical assays, and simulations provides a multi-scale perspective on lattice regulation.
- (4) The conceptual framework proposed lattice conformation as a determinant of post-translational modification accessibility is novel and potentially impactful for understanding

microtubule regulation.

Weaknesses:

There are a number of weaknesses in the paper, many of which can be addressed textually. Some of the supporting evidence is preliminary and would benefit from additional experimental validation and clearer presentation before the conclusions can be considered fully supported.

In particular, the authors should directly test in vitro whether Taxol addition can induce lattice exchange (see comments below).

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